ORACLE OS

Print pack — survival and rebuilding guides

English · Generated 2026-11-11 · 35

How to use and store these sheets

1. Print sections 1–3 first (first aid, water, the 50-year plan). Then the rest, in order.
2. Ordinary 80 gsm paper lasts decades if kept dry, dark and away from rodents. Use zip-lock plastic folders or doubled freezer bags.
3. Make at least two copies and keep them in different places (home, car, relative, shelter).
4. When the ink fades, hand-copy the most used chapters. Knowledge only survives if it is copied and taught.
5. In the grab bag carry only sections 1, 2, 4, 5 and 22–23, folded in a waterproof bag.
6. The full manuals (Where There Is No Doctor, FM 21-76, etc.) are in the 'manuais' folder. Print them double-sided.

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26. Communication without internet: AM, FM, long wave and two-way radio

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29. Protecting family, food and shelter in a crisis — security without violence

30. Looking after your mind in a crisis

31. Stretching and pain — back, neck, arms, legs, feet, cramps and heavy effort

32. Self-defence — Krav Maga, boxing, Muay Thai and jiu-jitsu (principles and core techniques)

33. Yoga — postures, breathing and relaxation (a safe 30–45 minute sequence)

34. Meditation — every major type, from ancient traditions to modern science

35. Chakras — origins, the seven centres, nadis, Kundalini and how to meditate with them

36. Perpetual calendar (2000–2100)

1. Emergency first aid — keeping someone alive until help arrives

Oracle OS guide. Based on IFRC 2025, European Resuscitation Council, WHO/ICRC "Basic Emergency Care" and ICRC "First aid in armed conflicts" (illustrated in the library: "IFRC International First Aid Guidelines", "US Army TC 4-02.1 First Aid", "MSF Clinical guidelines", "Hesperian Where There Is No Doctor"). Not a substitute for training — take a first aid course when you can.

Priorities

1. **Safety**: don't become another casualty (traffic, fire, electricity, gas, water, unstable structures).
2. **Call for help**: 999 or 112 (UK), 112 (Europe), 911 (US/Canada). No signal: send someone or use radio.
3. **Catastrophic bleeding** first, then **airway**, **breathing**, **circulation** (X-ABC).
4. Keep the casualty **warm**, calm and not alone. Note times and what you did.

Cardiac arrest — CPR

- Unresponsive and not breathing normally (or only gasping) = cardiac arrest. Call on speaker and send for a **defibrillator (AED)**.
- **Compressions**: centre of chest, hands stacked, arms straight, **5–6 cm** deep, **100–120 per minute**, let the chest fully recoil.
- If trained and willing: **30 compressions : 2 breaths**. Otherwise hands-only, without stopping.
- Switch the AED on as soon as it arrives and follow its voice prompts.
- **Drowning and children**: start with **5 rescue breaths**, then 30:2. Child: press 1/3 of chest depth (one or two hands). Infant: two fingers, about 4 cm.
- Continue until normal breathing returns, help takes over, or you are exhausted (swap every 2 minutes if possible).

Severe bleeding

- **Firm direct pressure** with a clean cloth for at least **10 minutes** without peeking. If it soaks through, add more on top.
- Deep wounds (groin, armpit, neck): **pack the wound** with gauze/cloth and press.
- **Tourniquet** only for life-threatening limb bleeding that pressure does not stop: 5–7 cm **above** the wound (not on a joint), tighten until bleeding stops, **write down the time**, do not release. Improvised: wide band (3–5 cm) and a stick to twist. Never thin cord.
- Lie them down and keep warm: risk of **shock**.

Shock

- Signs: pale, cold sweat, fast weak pulse, confusion, thirst.

- Lay flat, control bleeding, insulate above and below, nothing to eat or drink if surgery may be needed. Urgent help.

Choking

- Can cough → encourage coughing.
- Cannot cough, speak or breathe → **5 back blows** between the shoulder blades (lean forward), then **5 abdominal thrusts**. Alternate.
- **Infant**: 5 back blows (face down along forearm, head low) + 5 chest thrusts with two fingers. Never abdominal thrusts on infants.
- Becomes unresponsive → CPR.

Unresponsive but breathing

- **Recovery position**: on their side, top leg bent, head tilted back so the mouth drains. Check breathing every minute.
- Suspected spinal injury (fall, crash): keep the head in line; move only for danger or to open the airway.

Burns

- **Cool under cool running water for 20 minutes**, as early as possible (worth it up to 3 hours later). No ice, butter or toothpaste.
- Remove rings, watches and clothing that isn't stuck. Cover loosely with cling film or a clean plastic bag.
- Get help if: larger than the casualty's palm, face, hands, feet, genitals, joints, chemical or electrical burns, children and elderly.
- Chemicals: brush off dry powder, flush with lots of water 20+ min. Eyes: flush 15–20 min with the eyelid held open.

Fractures, sprains, dislocations

- Immobilise **in the position found**; splint across the joint above and below (sticks, magazines, cardboard, pillow). Check circulation (colour, warmth, feeling) after tying.
- Open fracture: cover with a clean cloth, don't push bone back, control bleeding.
- Sprain: rest, cold (wrapped), compression, elevation.

Cold and heat

- **Hypothermia** (shivering, confusion, slurred speech): shelter, remove wet clothes, insulate from the ground, blanket/sleeping bag, cover the head; warm sweet drinks **only** if fully alert. Severe (stops shivering, drowsy): handle very gently and horizontally.
- **Frostbite**: don't rub; rewarm in 37–39 °C water only if it will not refreeze.
- **Heat stroke** (hot, confused, may stop sweating): **cool immediately** — cold-water immersion is best; otherwise soak and fan, ice packs to neck, armpits, groin. Urgent help.
- **Heat exhaustion**: shade, lie down, water or oral rehydration solution.

Dehydration, diarrhoea, vomiting — home-made ORS (WHO)

- **1 litre safe water + 6 level teaspoons sugar + ½ level teaspoon salt.** Stir to dissolve. Should taste no saltier than tears.
- Small frequent sips, even with vomiting. Children keep breastfeeding/eating.
- Danger signs: no urine for 8+ hours, very drowsy, sunken eyes, blood in stool, baby not feeding → medical help.

Other common emergencies

- **Anaphylaxis** (swollen mouth/throat, breathing difficulty, rash): **adrenaline auto-injector** into outer thigh (through clothes), call 999/112, lie down legs raised (or sit up if breathing is hard). Second dose after 5 min if no better.
- **Asthma**: sit upright, blue reliever (salbutamol) — 1 puff every 30–60 s up to 10.
- **Heart attack** (chest pain spreading, sweating, nausea): sit, call, adult not allergic → **chew 300 mg aspirin**.
- **Stroke — FAST**: Face drooping, Arm weak, Speech slurred → Time to call; note the time.
- **Low blood sugar** (diabetic, confused, sweaty): if able to swallow, 15–20 g sugar (juice, sugar, sweets); repeat after 15 min.
- **Seizure**: clear the area, cushion the head, don't restrain or put anything in the mouth; recovery position afterwards. Over 5 minutes → help.
- **Poisoning**: don't induce vomiting; keep the packaging; burned mouth → small sips of water.
- **Snakebite** (adder in Europe): keep still, immobilise the limb, remove rings, hospital. No cutting, sucking or tourniquet.
- **Wounds**: irrigate with plenty of clean (or boiled and cooled) water, soap around it; remove dirt; cover. Infection (spreading redness, pus, fever) → antibiotics/doctor. Keep tetanus vaccination current.
- **Animal bites**: wash 15 minutes with soap and water (rabies risk); seek help.

Emergency childbirth

- Let it happen naturally; clean hands; support the baby's head, never pull.
- Dry the baby, skin-to-skin with mother, cover both. Not breathing: dry and rub the back; if still not, 5 breaths and infant CPR.
- No rush to cut the cord: wait 1–3 min, tie in two places and cut between with a clean/boiled blade. Placenta comes within 30–60 min; keep it to be checked. Heavy bleeding afterwards: rub the womb firmly through the belly and breastfeed.

Many casualties — rapid triage (START)

- **Can walk?** → GREEN (can wait).
- Not breathing after opening the airway → BLACK (dead/unsalvageable with current resources).
- Breathing over 30/min → RED (immediate).
- No radial pulse or capillary refill > 2 s → RED.

- Cannot follow simple commands → RED. Otherwise → YELLOW (delayed).
- Only stop to open airways and stop major bleeding; reassess constantly.

Minimum first aid kit

Gloves, gauze and bandages, plasters, tape, tourniquet, foil blanket, scissors, tweezers, saline, soap, paracetamol/ibuprofen, antihistamine, rehydration salts, thermometer, family's regular medication for 2+ weeks, allergy list.

2. Emergency drinking water — find, treat, store (English)

Oracle OS guide based on CDC, WHO, FEMA and IFRC guidance (see library: IFRC Household Water Treatment, WHO/WEDC notes, Practical Action SODIS).

How much

- At least **4 litres (1 gallon) per person per day** for drinking and cooking; more in heat, illness, pregnancy or hard work.
- Store at least **3 days** (72 hours); ideally 1–2 weeks. Remember pets.

Water already in your home

- Stored bottles; the **hot-water tank** (turn off power/gas, close the inlet, open the drain valve); the **toilet cistern** (tank, not bowl, and no coloured cleaners) — treat it; ice cubes; liquid in canned fruit and vegetables.
- House pipes: open the highest tap to let air in, collect from the lowest.
- **Never** drink untreated radiator, pool, central heating or flood water.

Treat it

1. **Settle and strain** cloudy water through a clean cloth or coffee filter (removes dirt only).
2. **Boil**: rolling boil for **1 minute** (3 minutes above 2,000 m). Most reliable against germs.
3. **Unscented household bleach** (5–6%): **2 drops per litre** (8 drops per gallon), stir, wait **30 minutes**. It should smell faintly of chlorine; if not, repeat and wait 15 more minutes. Double for cloudy or very cold water. Never use scented, gel or detergent bleach.
4. **Purification tablets**: follow the pack.
5. **SODIS**: clear PET bottles up to 2 L, clear water, direct sun for **6 hours** (2 days if cloudy).
6. **Portable filters** remove bacteria and protozoa; many do not remove viruses — add chlorine or boil if possible.

None of these remove fuel, chemicals or heavy metals. If water smells of fuel or looks oily or foamy, find another source.

Store it

Food-grade, clean, sealed containers away from sunlight and chemicals. Sanitise with 1 teaspoon of bleach per litre, rinse. Replace home-stored water every 6 months. Pour out — don't dip hands or cups in.

Rainwater

Clean roof and gutters, discard the first flush, cover the container, always treat before drinking.

Dehydration

Thirst, dark scanty urine, dizziness, dry mouth. Children: no tears, dry nappy. Give small frequent sips; for diarrhoea use oral rehydration salts (WHO home recipe if none: 1 litre safe water + 6 level teaspoons sugar + ½ level teaspoon salt).

3. The 50-year plan — surviving and rebuilding after a total collapse

Oracle OS guide. It is the "map" linking every other guide and manual in the library. Think of it in phases: priorities change in each. A phone won't last 50 years — Oracle's job is to **pass the knowledge on to people and paper** while it works.

Rules for every phase

1. **People first**: nobody lasts decades alone. Join trusted neighbours, share tasks, care for the most vulnerable.
2. **Water, shelter, fire, food, health** — in that order, almost always.
3. **Record**: write down on paper what works (seeds, dates, recipes, illnesses, births). Collective memory is the new "hard drive".
4. **Teach**: every skill should be known by at least 3 people.
5. **Oracle's power**: charge with a solar panel at midday; use power-saving mode; copy the library to another device and keep it switched off in a metal (Faraday) bag.

Phase 0 — first 72 hours

- Safety and the injured: "First aid", "Search and rescue".
- Water: store all the water you can now (baths, jerrycans, hot-water cylinder). See "Drinking water".
- Shelter and warmth: "Blackout, cold and shelter"; never braziers indoors.
- Information: FM/AM radio (Radio tab), coordinate with neighbours, meeting point.
- Inventory: food, medicines, tools, fuel, people and what each one can do.

Phase 1 — first month

- Secure water: source, filter, disinfection (boil, chlorine, SODIS), latrines 30+ m from wells and rivers.
- Food: ration stores, preserve perishables (salt, dry, smoke — "Food preservation"), fish, gather known plants (use "Identify a plant" but always confirm).
- Hygiene: wash hands with soap/ash, keep waste and faeces away from water — prevents cholera and dysentery (the big killers after disasters).
- Health: kit, essential medicines, mental health ("Mental health in a crisis", meditation, routine).
- Organisation: short daily meeting, watch rota, record of who needs what ("Community and organisation").

Phase 2 — first year

- **Sow**: choose open-pollinated seeds, prepare soil, compost, sowing calendar ("Seeds and grains", gardening books in the library). Save seed from the best plant of each harvest.

- **Animals**: chickens and rabbits (fast to breed), goats for milk ("Animals and veterinary care").
- **Store for winter**: dry, salt, smoke, cool cellar, preserves.
- **Durable shelter**: insulate, efficient (rocket) stove, chimney, watertight roof ("Natural building").
- **Long-term water**: protected hand-dug well, rainwater harvesting with a sand filter.
- **Health**: midwifery and newborn care ("Midwifery and medicinal plants"), teeth, wounds, isolating the sick.
- **Tools**: sharpening, handle repair, charcoal, first smithing from scrap.

Phase 3 — years 2 to 5

- Crop rotation (grains, legumes, roots, fallow), orchards and grafting, beekeeping (honey, wax, pollination).
- **Crafts**: smithing, pottery and kilns, carpentry, soap and candles ("Basic chemistry"), textiles (flax and wool: card, spin, weave), tanning hides and making shoes.
- **Off-grid energy**: water wheel/mill for grinding, dynamo to charge Oracle, animal traction ("Off-grid energy").
- **School**: teach reading, writing, counting, measuring, first aid and farming — every day ("Community and education").
- **Communication**: signals, Morse with light and mirror, messengers, hand-drawn maps ("Communication without power").
- **Rules and justice**: written agreements, conflict mediation, fair sharing.

Phase 4 — years 5 to 20

- Surpluses and trade with other communities: seeds, tools, cloth, salt, knowledge.
- Common weights, measures and calendar; records of property and births.
- Specialisation: smith, potter, midwife, teacher, healer, builder — with apprentices.
- Copy essential books by hand or print while paper lasts; create a community library.
- Recover simple technology: mills, lime kilns, bricks, crude glass, paper.

Phase 5 — years 20 to 50

- The generation born after the collapse takes over: it must have learned all this in practice.
- Writing, maths and basic science make it possible to rebuild the rest: chemistry, metallurgy, mechanics, medicine (LibreTexts, OpenStax and Wikipedia in the library are the "manual for rebuilding civilisation").
- Public health: clean water, vaccines if any exist, hygiene, nutrition — they triple life expectancy.

What to print now (while there is power)

First aid; drinking water; food preservation; seeds and grains; midwifery; fire and shelter; navigation; wild plants (dangerous look-alikes); soap; this plan. Use Oracle's

print button ("Print pack") and keep it in a waterproof bag.

What to buy/gather physically

USB solar panel 20–60 W + LiFePO4 power bank; spare phone with an Oracle copy in a Faraday bag; open-pollinated seeds; hand tools (axe, saw, spade, hoe, hammer, files, sharpening stone, needles, thread); cast-iron pots; medical kit; water filters; lighters and ferro rod; tarps; rope; paper and pencils.

4. Starting a fire safely (English)

Oracle OS guide (SurvivalManual / US Army FM 21-76 in the library).

Before lighting

Sheltered from wind, **away from trees, dry brush, roofs and tents**; clear a 3 m circle to bare soil; stone ring (avoid river stones — they can burst). Respect fire bans. Keep water, soil or sand ready.

Gather everything first

1. **Tinder** (catches a spark): cotton with petroleum jelly, paper, birch bark, dry grass, feather sticks, dryer lint, fine steel wool.
2. **Kindling**: dry sticks from match-thick to finger-thick (dead branches still on trees are drier).
3. **Fuel**: wrist-thick and bigger. Split wet wood — the inside is often dry.

Lays

Teepee (fast, cooking), log cabin (steady, good coals), star (push logs in as they burn; saves wood).

Ignition

Matches in a waterproof jar, lighter, **ferro rod** (scrape hard close to the tinder), **9 V battery + fine steel wool**, magnifying lens in sun, friction bow drill (practise first). **Never** use petrol.

Put it out

Spread embers, drown with water, stir, drown again until cool to the back of your hand.

Indoors

Only in a sound fireplace or wood stove. Never charcoal or camping stoves inside. Smoke and CO alarms.

5. Long blackout, cold and sheltering at home (English)

Oracle OS guide (FEMA, Ready Scotland, NHS, Nordic guides).

First hours checklist

Torch (avoid candles), check elderly/ill neighbours; unplug sensitive devices, leave one light on; keep fridge/freezer **closed**; fill containers while there is water pressure; turn on the radio; save phone battery (airplane mode, check at set times); keep cold-chain medicines in a cool box with ice (in-use insulin usually keeps weeks at room temperature — check the leaflet); have a plan for home medical equipment.

Staying warm without heating

- Everyone in **one small inner room**, other doors closed; cover windows with blankets/cardboard/plastic; block draughts with rolled towels.
- **Layers** of loose clothing, hat, wool socks, gloves; keep clothes **dry**.
- Sleep insulated from the floor; a blanket "tent" over the bed traps heat.
- Eat and drink warm drinks; **avoid alcohol**. Sealed hot-water bottles wrapped in cloth.
- **Never** heat the home with a gas cooker, oven, grill or generator (carbon monoxide).

Hypothermia

Signs: violent shivering, numb hands, slurred speech, confusion, drowsiness, clumsiness; babies: bright red cold skin, very low energy. Warm place, remove wet clothes, warm the **core first** (chest, neck, head, groin) with blankets or skin-to-skin; warm drinks **only if fully conscious**; don't rub the skin; call 112/999/911. Frostbite: rewarm in warm (not hot) water, don't rub, don't refreeze.

Extreme heat

Coolest room, shutters closed by day, open at night, drink regularly, wet the skin, avoid exertion; heatstroke (confusion, hot dry skin, collapse): cool immediately and call for help.

Generator

Outdoors only, at least 6 m from doors and windows; never back-feed the house wiring without a transfer switch.

6. Heating and cooking food without electricity or gas (English)

Oracle OS guide. The biggest killer in blackouts is **carbon monoxide (CO)** — invisible and odourless.

Golden rules

- **Never** use charcoal grills, camping stoves, generators or patio heaters indoors, in a garage, closed tent or enclosed balcony. Use them **outdoors**, away from windows and doors.
- CO symptoms: headache, dizziness, nausea, drowsiness, confusion → get to fresh air and call 112 / 999 / 911.
- A battery CO alarm is the best protection. Keep sand/water and a fire blanket near any flame.
- **Never heat a sealed can** — it can explode. Open it or pour into a pan.

No-cook food

Canned beans, fish, vegetables and fruit (already cooked, safe cold), bread, crackers, peanut butter, nuts, cereal with UHT milk, washed raw fruit and vegetables.

Ways to heat

- **Camping gas stove** outdoors or on an open balcony, lid on the pot. A 230 g cartridge gives roughly 1–1.5 hours of flame.
- **Outdoor wood fire** in a pit or stone ring away from trees, roofs and strong wind (see "Starting a fire").
- **Fireplace or wood stove** only if the chimney is sound and normally used. Never burn plastic or treated wood.
- **Retained-heat ("haybox") cooking**: boil rice, beans or soup 5–10 min, cover tightly, wrap the pot in blankets/pillows in a box. It finishes cooking in 1–3 h (soaked beans 3–5 h), saving up to 70% fuel.
- **Solar oven**: foil-lined box, clear lid, dark covered pot; strong sun, 2–4 h.
- Candles and "flowerpot heaters" barely heat and start fires — avoid.

Save fuel

Soak pulses overnight, cut food small, always use a lid, little water, cook several things at once, reuse hot water for washing.

Kitchen hygiene

Wash hands; wash raw food with treated water; eat cooked food promptly — without a fridge, leftovers no more than 2 hours at room temperature (1 hour above 32 °C).

7. Making food last — without a fridge and long term (English)

Oracle OS guide (USDA/FoodSafety.gov, FAO, USDA Complete Guide to Home Canning in the library).

During a power cut

- Keep fridge and freezer **closed**: fridge ≈ **4 hours** cold; full freezer ≈ **48 hours** (half full ≈ 24 hours).
- Eat fridge food first, then freezer, then tins.
- Discard meat, fish, eggs, dairy, cooked leftovers and sauces kept **above 4 °C (40 °F) for over 2 hours**. When in doubt, throw it out — never taste to test.
- Frozen food that still has ice crystals or is at 4 °C or below can be refrozen.

Long-life pantry (rotate: oldest first)

Rice, pasta, flour, oats, dried beans/lentils/chickpeas, tins, oil, salt, sugar, honey, powdered/UHT milk, tea/coffee, nuts, crackers, peanut butter, dried fruit, vitamins. Cool, dry, dark, sealed against damp, insects and rodents.

Traditional preservation

- **Drying** fruit, vegetables, herbs, thin strips of lean meat in sun and wind or dry heat, under insect netting.
- **Salting** fish/meat in coarse salt, then drying; soak before cooking.
- **Smoking**: low heat + smoke + drying; use non-resinous hardwood.
- **Fermenting**: cabbage/vegetables with **2–3% of their weight in salt**, packed under brine, 18–22 °C for 1–4 weeks; should smell sour, not rotten; discard coloured mould, slime or foul smell.
- **Pickling** in vinegar of at least 5% acidity; **sugar** (jams).
- **Natural cold**: cellar or pit for potatoes, carrots, apples, squash, onions; **zeer pot** (pot in pot with wet sand) cools by evaporation in dry climates.
- **Fat**: cooked meat sealed under lard (confit) keeps for weeks somewhere cool.

Home canning — botulism warning

High-acid foods (fruit, acidified tomatoes, pickles, jams): boiling water bath. **Low-acid foods (vegetables, beans, meat, fish, soups) only in a pressure canner** — a water bath does not kill *Clostridium botulinum*. Use tested times (USDA guide). Discard bulging, leaking, spurting or foul-smelling jars without tasting.

Signs of spoilage

Sour or rotten smell, slime, mould, swollen/seam-dented/rusty cans, domed lids.

8. Personal hygiene and bathing with little water (English)

Oracle OS guide (WHO, Sphere, Hesperian). Handwashing is the single most important measure.

Handwashing

- Soap and water, scrub **20 seconds** (palms, backs, between fingers, nails, wrists).
- After the toilet, before cooking and eating, after nappies, rubbish, animals, and caring for the sick.
- No running water: build a **tippy-tap** (hanging jug with a small hole, tilted by a stick and string; soap on a string).
- Hand sanitiser with **at least 60% alcohol** when hands are not visibly dirty. No soap: clean ash rubbed and rinsed is better than water alone.

Sponge bath with 2–4 litres

Basin, cloth, soap. Order: **face → neck and arms → torso → legs → feet → genitals last**. Rinse with a wrung cloth, change water if very dirty, dry well between toes and skin folds.

Teeth

Brush twice a day with treated or boiled-and-cooled water.

Menstrual hygiene

Pads, tampons, cup or clean cotton cloths. Wash reusable cloths with soap (cold then hot water) and **dry in the sun**. Boil cups between cycles.

Toilet without water (bucket toilet)

- **Two buckets**: urine (dilute and pour away from wells and streams) and faeces.
- Line with a strong bag; after each use cover with **sawdust, dry soil, ash or lime**; lid on. Tie off half-full bags, store in a closed bin away from the house and water until collection.
- **Outdoor latrine**: at least **30 m** from any well, stream or water source, and downhill of it.
- Don't use a flush toilet without water — it blocks and contaminates.

Laundry and skin

Wash in a basin with a little soap, rinse once, dry in the sun. Clean small wounds with treated water and soap and cover; seek help for spreading redness, pus, fever or red streaks.

9. War, attacks, nuclear explosion and chemical incidents (English)

Oracle OS guide (FEMA, CDC, Sweden's "In case of crisis or war", Norway/Finland guides in the library). Always follow official instructions on radio and alerts.

GO IN, STAY IN, TUNE IN

Get into the nearest solid (brick/concrete) building, stay inside away from windows, tune to public radio (UK: BBC Radio 4 / BBC Radio Scotland — see Radio screen).

Sirens and alerts

Meanings vary by country. On hearing one: go inside, close doors/windows, turn on the radio. Keep phone emergency alerts enabled.

Air raid / shelling

Public shelter, basement, underground car park or metro; at home the innermost windowless room on the lowest floor. Put two walls between you and outside. Stay away from glass. Do not film troops or strike sites; do not post locations online.

Nuclear explosion

- **Don't look at the flash.** Outdoors: lie down, cover your head.
- You have **minutes** before fallout arrives: **get inside** the best shelter — centre of a concrete building or basement.
- **Stay inside at least 24 hours** unless told otherwise; fallout radiation drops sharply over the first hours and days.
- If exposed: **remove outer clothing** before entering (removes up to ~90% of contamination), bag it away from people, shower with soap and shampoo, **no conditioner**; blow your nose, wipe ears and eyelids.
- Sealed food and water are safe — wipe the packaging first.
- **Potassium iodide (KI)** protects only the thyroid and should be taken **only when health officials say so**.

Chemical release

Go inside, close windows, turn off ventilation, seal gaps with wet towels and tape; many toxic gases are heavier than air — go **upstairs**. Outdoors: move crosswind and upwind. Skin contact: remove clothing, wash with lots of soap and water, flush eyes 15 minutes.

Evacuation

72-hour grab bag: water, food, medicines and prescriptions, ID/documents (copies on phone too), cash, charger and power bank, battery radio, torch, warm clothes, first-aid kit, paper list of contacts. Agree a meeting point near home and one outside town. Take pets if you can.

Checkpoints and afterwards

Carry ID; obey checkpoints; show your hands; don't run. Don't return home until allowed. Beware damaged buildings, gas leaks (no flames or switches), fallen power lines and unexploded ordnance — **never touch military objects**; move away and report.

10. Search and rescue — rubble, fire, floods and lost people

Oracle OS guide. Based on FEMA CERT, INSARAG (UN), the US&R Field Operations Guide and RNLI (illustrated in the library: "FEMA CERT Basic Training", "INSARAG Guidelines vol. II", "US Urban SAR Field Operations Guide", "NWCG Incident Response Pocket Guide", "PHMSA Emergency Response Guidebook").

Golden rule

Don't become another casualty. Many would-be rescuers die trying to save someone. Assess first, act second, always keep an exit.

Size up the scene (before going in)

- What happened? How many people? Where could they be?
- Hazards: fire, smoke, gas (smell → get out, don't touch switches), electricity, water, chemicals (orange plates with numbers: look them up in the "Emergency Response Guidebook"), collapsing structures, aftershocks.
- Resources: who is with you, tools, light, radio, remaining daylight.
- Shut off gas and electricity at the meter if safe.

Damaged buildings (earthquake, explosion, collapse)

- **Light damage** (cracks, glass): enter carefully.
- **Moderate damage** (cracked walls, damaged stairs): only quick entry for visible casualties.
- **Heavy damage** (fallen walls/roof, leaning, off foundation): **do not enter**. Mark it, cordon it, wait for professionals.
- Work in pairs, with helmet/gloves/dust mask and sturdy shoes. One person stays outside and knows where you are.

Finding casualties

- **Call out and listen**: "If you can hear me, knock 3 times!" Then **total silence** for 1–2 minutes. Repeat from several points and triangulate.
- Search void spaces: under tables, along interior walls, stairwells, corners, vehicles.
- Surface and lightly trapped casualties first (most survivors are pulled out in the first hours by neighbours).
- **Marking** at the entrance (spray or chalk): an X — left: team; top: date and time; right: hazards; bottom: casualties (alive/dead). Prevents repeated searches.

Freeing someone trapped

- **Lever and crib**: lift a little with a lever (bar, strong plank on a fulcrum), slide in a block; **lift an inch, crib an inch**. Never leave a load unsupported.
- Don't pull beams that hold other parts up. Remove debris by hand, passing it along a chain, top down.

- **Crush syndrome**: someone with a limb crushed for more than 1–2 hours can deteriorate suddenly when freed. Give water if conscious, get medical help, monitor; don't leave them alone.
- Protect the casualty from dust (cloth over face) and cold during extraction.

Moving casualties

- Only for immediate danger. Suspected spinal injury: move as a unit, head in line.
- **Clothing drag** (by the shoulders of the top, head cradled on your forearms) or **blanket drag**.
- **Two-person carry**: hand seat, or one at the armpits and one at the knees.
- **Improvised stretcher**: two 2 m poles + coats with sleeves turned inside, or a blanket folded in thirds around the poles. Test with a healthy adult first.

Fire and smoke

- Crawl below smoke; feel the door with the back of your hand before opening — hot, don't open.
- Close doors behind you (slows the fire). Trapped: close the door, seal gaps with wet cloth, signal from the window.
- Clothes on fire: **stop, drop, roll**. Cool burns 20 min with water.
- Carbon monoxide (generators, barbecues indoors): headache, nausea, drowsiness in several people → fresh air now.
- Wildfire: move to already-burnt ground, bare ground, water or a wide road; never run uphill ahead of the fire.

Water — floods, rivers, ice

- **Reach, throw, row, go** — in that order. Reach with a stick, rope, clothing; throw something that floats (empty jerrycan, ball, life jacket). Only go in trained and with buoyancy.
- Moving water: **15 cm can knock an adult over, 30 cm can float a car**. "Turn around, don't drown."
- Ice: don't run to the casualty. Lie flat, spread your weight, push out a pole/ladder; pull them to the bank.
- See the "Water, rivers and sea" guide for currents, crossings and sea survival.

Signals to be found

- **3 of anything** = distress: 3 whistle blasts, 3 fires in a triangle, 3 flashes. **SOS** in Morse: · · · — — — · · ·
- A whistle carries much further than a voice and uses less energy.
- **Ground-to-air signals** (large, 3 m+, contrasting): **V** = need help; **X** = need medical help; **N** = no; **Y** = yes; **arrow** = went this way.
- Mirror/shiny surface to flash sunlight at aircraft; bright colours; smoke by day (green leaves), fire by night.

- Radio: marine VHF channel 16; PMR446 channel 8 (code 0) and channel 1 are common emergency channels; say who, where, what, how many.

Lost person

- If it's you: **STOP** — Stop, Think, Observe, Plan. Stay put if someone knows your route; shelter, signal.
- Children: teach them to "hug a tree" and stay still.
- Searching: start at the last seen point; check likely routes, junctions, water, shelters. Line search with people within sight of each other, calling and pausing to listen. Mark areas already searched.

Reporting an emergency (radio or phone)

1 Who you are and where you're calling from. 2. **Where** (Oracle coordinates: Emergency tab). 3. What happened. 4. How many casualties and their state. 5. Hazards. 6. What you've done and what you need. Don't hang up first.

Household rescue kit

Helmet or hard hat, work gloves, goggles, FFP2 masks, head torch + batteries, whistle, crowbar, hammer, saw, gas/water shut-off key, 15–30 m rope, barrier tape, chalk/spray, markers, blanket, first aid kit.

11. Childbirth, mother-and-baby care, and medicinal plants

Oracle OS guide. Manuals in the library: Hesperian "A Book for Midwives", "Where There Is No Doctor", WHO "Pregnancy, Childbirth, Postpartum and Newborn Care", MSF Obstetrics, Culpeper "Complete Herbal", WikiMed/MDWiki. This guide does not replace a trained midwife: train at least two people per community.

Pregnancy — follow-up

- Varied food (pulses, eggs, dark greens, fish), clean water, rest. Avoid alcohol, smoking and abortifacient herbs (rue, pennyroyal, mugwort, tansy).
- Measure the belly monthly: the top of the womb rises ~1 cm per week from 20 weeks (navel ≈ 20–22 weeks).
- **Danger signs in pregnancy** — get the most trained person now: vaginal bleeding; bad headache, blurred vision, sudden swelling of face/hands or fits (pre-eclampsia/eclampsia); fever; waters breaking before 37 weeks; baby stops moving; strong constant belly pain.

Prepare for the birth (clean kit)

Soap, boiled water, clean boiled cloths, 2 boiled ties/strings, a new razor blade or scissors boiled 20 min, gloves or washed hands and short nails, a blanket for the baby, a container for the placenta, a torch. **The "3 cleans": clean hands, clean surface, clean cord cut.**

Labour (normal)

1. **Opening**: regular contractions getting stronger; let the mother walk, drink, pee, choose her position. First baby: often 12 h or more.
2. **Pushing**: urge to push; support the perineum with a cloth, let the head come out slowly. If the cord is loose around the neck, slip it over the head.
3. Baby out: dry, skin-to-skin on the mother's chest, cover. If not breathing: dry and rub the back, clear mouth/nose, head slightly back, 5 gentle breaths covering nose+mouth, then ~40 breaths/min; if the heart is below 60 after 1 min of breaths, two-thumb compressions (3:1).
4. **Cord**: wait 1–3 min (or until it stops pulsing), tie in two places (2 and 5 cm from the baby), cut between them with a clean blade. Keep it dry, put nothing on it.
5. **Placenta**: normally comes within 30 min. Don't pull hard on the cord. Then rub the womb (it should feel hard like a ball) and put the baby to the breast — both reduce bleeding. Check the placenta is complete.

Birth emergencies — recognise

- **Postpartum haemorrhage** (> 500 ml, or the mother turns pale, dizzy, fast pulse): continuous womb massage, baby at the breast, empty the bladder, legs raised, fluids; if misoprostol/oxytocin is in the kit, use as the manual says. Bimanual compression (midwives' manual). It is the #1 cause of maternal death.

- Pushing stage > 2 h without progress or labour stalled > 12 h; transverse baby; arm or cord coming first; difficult breech; fever with bad smell; fits. See the specific chapters in the manual.

After the birth and the newborn

- Breastfeed in the first hour and on demand (8–12 times/day); colostrum is precious. Keep the baby warm (skin-to-skin, hat).
- Watch the mother for 6 weeks: fever, foul-smelling blood, belly pain → infection (see Where There Is No Doctor).
- Danger signs in the baby: not feeding, cold or feverish, fast (> 60/min) or grunting breathing, yellow skin within 24 h, red/pus-filled navel, fits.
- Space pregnancies ≥ 2 years: exclusive breastfeeding protects only in the first 6 months and while periods have not returned (LAM method).

Medicinal plants — prudent use

Rules: identify with certainty (2 sources, see the Plants module), start with low doses, one plant at a time, nothing in pregnancy without certainty, children half or less. Teas: 1 tablespoon dried herb (or 2 fresh) per cup; infusion for leaves/flowers (10 min covered), decoction for roots/bark (boil 10–15 min).

- **Chamomile**: digestion, mild colic, gentle calming.
- **Peppermint**: nausea, gas (avoid with strong reflux and in babies).
- **Ginger**: nausea (including morning sickness at food doses), colds.
- **Garlic**: mild antimicrobial in food; raw crushed on wounds can burn — dilute.
- **Plantain (Plantago)**: chewed/crushed leaf on stings, small wounds; tea for cough.
- **Calendula**: salve/oil for skin, superficial wounds, nappy rash.
- **Yarrow (Achillea)**: stops minor bleeding in wounds (not in pregnancy).
- **Willow bark**: pain and fever (contains salicylates) — not for children, aspirin allergy or ulcers.
- **Thyme**: cough, sore throat (gargle).
- **Elder (flowers; berries only cooked)**: colds, fever.
- **Echinacea**: start of colds (short term).
- **Honey**: cough (never babies < 1 year), wounds (pure honey as a dressing).
- **Valerian**: insomnia, anxiety.
- **Comfrey**: external only, on bruises — toxic to the liver if swallowed.
- **Dangerous** look-alikes: hemlock (looks like wild carrot/parsley), foxglove, deadly nightshade, monkshood, yew. If in doubt, don't use it.
- Dry in the shade, store in dark labelled jars with name and date (keeps ~1 year).

12. Wild plants — identifying, eating, using and avoiding poisoning

Oracle OS guide. Based on US Army FM 3-05.70 / ATP 3-50.21 (edible and poisonous plants, **illustrated** in the library), offline Wikipedia and Hesperian. Use **Identify a plant from a photo** (Library tab or Home → Plants) as a starting point, never as final confirmation.

Golden rules

1. **If in doubt, don't eat it.** Hunger kills in weeks; some plants kill in hours.
2. Confirm with **two sources** (photo + illustrated guide + someone who knows). Check **every part**: leaves (shape, arrangement, smell), stem (hollow? spotted? hairy?), flower, fruit, root.
3. **Mushrooms: never** from a photo or a "test". Several deadly ones (Amanita phalloides) look edible and symptoms come too late.
4. Avoid plants by roads, sprayed fields, polluted water and industrial land.
5. Children, pregnant women, the elderly and the sick are more sensitive.

Warning signs (avoid unless absolutely sure)

- Carrot family (**Apiaceae**: white umbrella-shaped flowers) — includes **hemlock and hemlock water dropwort**, the deadliest plants in Europe. Purple-blotched stem and mousy smell = hemlock.
- Milky sap (except dandelion and a few known ones), bitter-almond smell, leaves in threes (poison ivy in America), seeds in pods (many wild legumes are toxic), white or yellow berries, almond-like fruits.
- Strong bitterness or burning in the mouth → spit it out.

Classic dangerous look-alikes (Europe)

- **Wild garlic** (smells of garlic when crushed) vs **lily of the valley**, **autumn crocus** and **lords-and-ladies** (no garlic smell, deadly).
- **Wild carrot / parsnip** vs **hemlock**, **hemlock water dropwort** (Oenanthe — deadly, in ditches and riverbanks).
- **Comfrey** vs **foxglove** (digitalis, affects the heart).
- **Wild parsley/chervil** vs **fool's parsley** (Aethusa).
- **Elder** (ripe berries edible cooked; raw berries, leaves and stems toxic) vs **dwarf elder** (toxic).
- **Bilberries** vs **deadly nightshade** berries (shiny black with a 5-pointed "star" of sepals — deadly).

Common useful plants in the UK and Portugal (always confirm)

- **Nettle**: young leaves cooked (boiling removes the sting) — soup, tea; fibre for cordage.
- **Dandelion**: young leaves in salad (bitter), roasted root as "coffee", flowers.

- **Bramble/blackberry**: raw fruit; young leaves for tea.
- **Elder**: flowers for infusion/cordial; ripe berries **cooked only**.
- **Plantain**: young leaves cooked; crushed leaf on insect stings.
- **Clover**, **sorrel** (small amounts — oxalates), **purslane**, **wild mustards**.
- **Sweet chestnut** (boiled/roasted; don't confuse with **horse chestnut**, whose husk has short sparse spikes and nut has no point — toxic), **hazelnuts**, **acorns** (soak and boil in several changes of water to remove tannins), **stone-pine nuts**.
- **Rose hips**: rich in vitamin C, remove the hairy seeds.
- **Seaweed** from clean water: sea lettuce, laver/nori.

Universal edibility test (last resort only, never for mushrooms)

Takes about 24 hours per plant part. Only test abundant plants.

1. Fast for 8 hours (water only). Separate the plant into parts (leaf, stem, root, flower, fruit) and test **one part at a time**.
2. Smell: strong unpleasant, bitter-almond or peach smell → reject.
3. Rub on the wrist or inner elbow; wait 15 min. Burning, itching or redness → reject.
4. Prepare it as you would eat it (cooking is safer). Touch to the lip 3 min; then the tip of the tongue 15 min; then chew and hold in the mouth 15 min **without swallowing**. Any burning, numbness or bitterness → spit and reject.
5. Swallow a small amount and wait **8 hours** eating nothing else. Nausea, cramps → drink water, activated charcoal if you have it; inducing vomiting is generally not recommended.
6. No reaction: eat ¼ cup prepared the same way and wait another 8 hours. No reaction → that part, prepared that way, is probably safe.

Uses beyond food

- **Firewood**: birch (bark burns even damp), pine (resin = fire starter), oak and ash (long-lasting embers). Don't burn **yew**, **oleander** or **poison ivy** (toxic smoke).
- **Cordage/fibre**: nettle, inner bark of lime/willow, cattail (see "Ropes and knots").
- **Shelter**: conifer boughs for an insulating bed; dry bracken.
- **Traditional remedies** (variable effect, not a substitute for medicine): willow bark (salicylates, pain — avoid in children and with blood thinners), plantain (stings), chamomile and lime blossom (calming tea), thyme (cough). Doses and interactions: see "MSF Essential drugs" and "Where There Is No Doctor".
- **Insect repellent**: smoke from green leaves; crushed myrtle and eucalyptus.
- **Water**: tree leaves in a plastic bag in the sun (transpiration) give a little water; never from toxic plants.

Plant poisoning — what to do

- Remove remains from the mouth, keep a sample of the plant (for identification), call 999/112/NHS 111 or the poisons centre (Portugal **800 250 250**).
- Don't induce vomiting. Small sips of water. Recovery position if drowsy. CPR if breathing stops.

- Sap on skin (e.g. **giant hogweed** causes burns in sunlight): wash at once with soap and water, keep out of the sun for 48 hours.

13. Fishing to survive — rivers, lakes and sea

Oracle OS guide. Based on the FAO "Fisherman's Workbook" and FAO Training Series, US Army FM 3-05.70/21-76 (fishing chapters) and MSF/WHO (food safety) — in the library **with drawings of hooks, knots, traps and nets**. In normal times obey the law: fishing licences (England/Wales: Environment Agency rod licence; Scotland: salmon/trout permissions; Portugal: recreational fishing licence), minimum sizes and seasons. Nets, set lines and traps are usually illegal outside an emergency.

Where and when fish are

- **Rivers**: slack water behind rocks, undercut banks, shade under trees, where two streams meet, the tail of deep pools, near bridges.
- **Lakes**: weed beds, jetties, drop-offs, stream inlets; early morning and evening in the shallows, midday deeper.
- **Sea**: rocks and piers, beaches with gentle surf (fish just behind the breakers), estuaries on a rising tide, channels between sandbanks.
- Best times: **dawn and dusk**, before storms, at the turn of the tide.

Improvised gear

- **Line**: kit fishing line, dental floss, paracord inner strands, sewing thread, twisted bark fibres (see "Ropes and knots").
- **Hook**: safety pin, bent and sharpened wire, thorns (hawthorn, bramble), carved bone or hardwood, can ring-pulls.
- **Gorge hook**: 2–3 cm stick or bone sharpened at both ends, tied in the middle; the fish swallows it with the bait and it jams crosswise.
- **Bait**: worms (dig damp soil, under stones), grubs, insects, crickets, fish guts and eyes, mussels, kneaded bread, sweetcorn. Predators hit shiny moving lures (scrap of tin, feather).
- **Float**: dry stick, cork, thick feather. **Weight**: small stone or nut.

Methods

- **Handline**: line wound on a stick; cast with the wrist, feel for bites and strike firmly.
- **Improvised rod**: flexible 2–3 m pole (willow, hazel), line tied at the tip and along the pole.
- **Set lines** (emergency): several hooks on a line strung between banks or from a springy branch; check at dawn and dusk.
- **Bottle/funnel trap**: large bottle cut, neck inverted inside, bait inside, weighted; small fish and crabs get in and can't get out. Bigger version woven from willow in a cone.
- **Stone fish trap / tidal weir**: semicircle wall of stones on a beach or shallow stream with a narrow entrance; the falling tide traps fish.
- **Net** (emergency): hand net (shirt/cloth on a branch hoop) for small fish in shallows; gill nets per the FAO instructions in the library.

- **Spear**: pole with a sharpened forked tip, in shallow still water; aim below the fish you see (water bends the image).
- Night fishing with a light attracts fish and squid.

Shellfish and other water food

- **Mussels, cockles, clams**: only from clean water, away from sewage and harbours; **never during "red tides"/algal blooms** or when warnings are posted — the toxins survive cooking. Must be alive (close when tapped). Cook until open; discard those that stay shut.
- **Crabs and crayfish**: baited drop net or by hand under stones; cook thoroughly.
- **Seaweed** (sea lettuce, laver/nori, kelp): from clean water, attached to rocks (not loose on the beach); wash well.
- Avoid: pufferfish (poisonous), unfamiliar spiny or scaleless reef fish in the tropics (ciguatera), fish found dead, fish from polluted water or near outfalls.

Preparing fish

1. Kill it straight away (sharp blow to the head) — keeps better.
2. **Scale** (tail to head), **gut**: slit the belly from vent to gills, remove guts and gills, rinse.
3. Fresh fish: bright eyes, red gills, firm flesh, sea smell. Cloudy eyes, brown gills, soft flesh or strong odour → **throw away**.
4. **Always cook freshwater fish thoroughly** (parasites): until the flesh is opaque and flakes off the bone. Skewered over embers, wrapped in leaves/clay, or in soup (uses everything).
5. Guts make bait or fertiliser; bury them away from camp (they attract animals).

Preserving

- **Smoking**: thin fillets over cool hardwood smoke (oak, beech, fruit wood), 1–2 days to preserve.
- **Drying**: thin fillets in sun and wind, protected from flies, until hard.
- **Salting**: layers of fish and coarse salt; then dry. See "Food preservation".

Water safety while fishing

Unstable banks, slippery rocks and freak waves kill anglers every year. Life jacket, never alone on rocks, check the tide table, watch the sea for 10 minutes before going down. See "Water, rivers and sea".

14. Hunting, trapping and using animals

Oracle OS guide. Manuals in the library: FM 21-76 (traps and snares), "Camp and Trail Methods", "The Trapper's Guide", "Fishing" (Oracle guide), "Animals" (Oracle guide). In normal times, follow hunting and animal-welfare laws; traps are for food only, checked every day, never on paths people use.

Principles

- Small game (rabbit, hare, squirrel, birds, fish) gives more calories for the effort than big game. Fishing and traps work while you do something else.
- **Read sign**: runs (beaten paths through grass), tracks, fresh droppings, burrows, feeding and drinking spots. Set traps on runs between burrow and food/water, at narrow points.
- **Many traps** (10–20) raise the odds; check at dawn and dusk. Minimise human scent (gloves, smoke parts over a fire, don't over-handle).
- Never leave traps abandoned.

Simple traps (details and drawings in FM 21-76)

- **Simple snare**: fine wire or strong cord with a running noose; ~10 cm loop for rabbit, 10 cm off the ground (a fist), on the run, anchored to a stake or firm branch. Small sticks guide the animal into the loop.
- **Spring snare**: noose tied to a bent sapling held by a trigger; when tripped it lifts the animal.
- **Deadfall (figure-4)**: flat stone/log propped on a baited 3-stick trigger; for small rodents.
- **Bird traps**: snares on perch branches; upturned baskets with a trigger stick.
- **Fish trap**: funnel basket of willow or a cut bottle; stone weirs in streams (see Fishing).

Hunting

- **Self bow**: yew, ash, elm, hazel or willow 1.5–1.8 m, season for weeks; fibre/sinew string. Arrows from straight shoots, bird feathers, a sharpened fire-hardened tip, bone or metal. Practise a lot before you need it.
- **Sling**, **spear**, **throwing sticks** for birds and small game.
- Stalking: downwind, slowly, freeze when the animal looks, use cover. Best times: dawn and dusk.
- Kill quickly and humanely; never wound without following up.

Preparing the meat (safely)

1. Bleed at once (head down, cut the neck vessels).
2. **Gut**: shallow cut from breastbone to pelvis without piercing the intestines or bladder; remove the organs; keep heart, liver (if healthy — no spots/cysts) and kidneys.

3. **Skin**: rabbits — cut round the hind legs and pull the skin off like a sock; larger animals — cuts down the legs and centre line, separate with the knife.
4. Cool the carcass in the shade, away from flies (cloth/bag).
5. **Cook thoroughly** (rabbit, wild boar, bear: risk of tularaemia and trichinosis). Gloves when skinning wild rabbits; don't eat animals that seemed sick or sluggish.
6. Don't live on lean meat alone (rabbit) for weeks — "rabbit starvation": add fat and carbohydrates.

Use everything

- Hide: tan it (see "Textiles and leather").
- Sinew: dry and shred for bowstrings and sewing.
- Bones: needles, fish hooks, tools, broth; bone ash as fertiliser (phosphorus).
- Fat: render into tallow (candles, soap, preserving).
- Preserving meat: dry in thin strips in sun/wind or cold smoke (jerky), salt, or hot-smoke.

15. Seeds and grains — from seed to bread, and saving seed for next year

Oracle OS guide. Manuals in the library: FAO (community seeds, seed production), "Home Vegetable Gardening", "The Vegetable Garden", "War Gardens", "Ten Acres Enough", Practical Action (irrigation, pumps), Gutenberg agriculture (illustrated, in Encyclopedias).

1. Which seeds to keep

- **Open-pollinated / heirloom** (not "F1" hybrids): only these breed true. Label each batch with name, year and place.
- Grains: wheat, rye (tolerates poor soil and cold), barley, oats, maize. Legumes: beans, peas, broad beans, chickpeas, lentils (fix nitrogen). Roots: **potatoes** (most calories per m²), sweet potato (warm climates), Jerusalem artichoke. Squash, cabbages, onion, garlic, carrot, beetroot, tomato, pepper. Herbs: parsley, coriander, thyme.
- Store in dry, cool, dark jars; typical life 2–5 years (onion 1–2, tomato and beans 4–6). Germination test: 10 seeds on damp paper; how many sprout in 7–10 days = germination %.

2. Soil fertility without chemical fertiliser

- **Compost**: layers of green (kitchen scraps, grass) and brown (dry leaves, straw) 1:2, damp like a wrung sponge, turn every 2–3 weeks; ready in 3–6 months.
- Well-rotted **manure** (chicken, cow, horse); never fresh on leaves eaten raw.
- **4-year rotation**: legumes → leafy (brassicas) → roots → potatoes/grain; prevents disease and soil exhaustion.
- **Mulch** (straw, leaves): keeps moisture, smothers weeds.
- **Green manure**: sow clover/field beans/lupins and dig in before flowering.
- Wood ash (potassium, small amounts), crushed eggshells (calcium).

3. Calendar (northern hemisphere, temperate)

- **Winter** (Dec–Feb): plan, repair tools, sow under cover (onions), prune fruit trees.
- **Spring** (Mar–May): potatoes (Mar–Apr), peas, broad beans, carrots, lettuce, cabbages; after frosts: beans, maize, squash, tomatoes.
- **Summer** (Jun–Aug): water early morning, harvest, save seed, sow winter brassicas.
- **Autumn** (Sep–Nov): **winter wheat and rye** (Oct–Nov), garlic, broad beans; harvest squash and potatoes; store.
- Southern Portugal ~3–4 weeks earlier; Scotland 2–4 weeks later.

4. From grain to bread

1. **Harvest** when the grain is hard and the straw yellow (barely dents with a fingernail). Cut with sickle/scythe, tie in sheaves, dry 1–2 weeks in stooks.
2. **Thresh**: beat the sheaves with a flail on a clean cloth or floor.

3. **Winnow**: toss the grain in a light breeze; chaff blows away, grain falls.
4. **Store** in sealed containers against rats and damp (drums, lidded clay pots); dry grain keeps for years.
5. **Grind**: hand quern (two flat stones, the upper one turns), water/wind mill, or mortar. Sieve for finer flour.
6. **Sourdough starter**: flour + water 1:1, stir and "feed" daily for 5–7 days until bubbly and sour-smelling; keep a little from each dough for the next.
7. **Bread**: 1 kg flour + 600–700 ml water + 20 g salt + 200 g starter; knead 10 min, rise 4–12 h, shape, bake in a wood oven (~220–250 °C) 40–50 min. See "A Treatise on Bread" in the library.
8. **Maize**: nixtamalise (cook with lime or wood ash) to prevent pellagra if it is a staple.

5. Saving seed from your own plants

- Pick the **best plants** (healthy, productive, survivors) and let them ripen fully.
- **Dry seeds** (beans, peas, grains, lettuce, onion): dry on the plant, thresh, dry 1–2 more weeks in shade.
- **Wet seeds** (tomato, squash, cucumber): scoop out; tomato ferments 2–3 days in water, then rinse and dry.
- Keep varieties of the same species that cross (maize, squash, brassicas) apart by distance or timing.
- Potatoes and garlic: keep the best tubers/bulbs for planting.

6. Watering

Drip irrigation from punctured bottles, buried clay pots (ollas), rainwater harvesting, water at dusk or dawn. See Practical Action (micro-irrigation, treadle pump).

16. Animals — raising, feeding, treating and using them

Oracle OS guide. **Illustrated** manuals in the library: CVM "Raising Healthy" (goats, rabbits, poultry, pigs, sheep, cattle) and "Drugs and their usage", Peace Corps and FAO (beekeeping), Langstroth, "Poultry", "Sheep, Swine, and Poultry", "Pratt's Practical Pointers", "Domestic Animals"; cheese and bread in "Seeds and grains".

Where to start

1. **Chickens**: daily eggs, meat, eat scraps and insects, excellent manure. Coop shut at night (foxes), 4–5 m² of run per bird, nest box, perch, clean water. A broody hen sits ~21 days; keep a cockerel for chicks.
2. **Rabbits**: fast breeding (gestation ~31 days, 6–8 kits), eat grass and garden scraps, lean meat. Dry shaded hutches; keep bucks apart.
3. **Goats**: milk (1–3 L/day), meat, hide; browse scrub. Strong fences; goats need company. Gestation ~150 days.
4. **Bees**: honey (sugar and antiseptic), wax (candles, waterproofing), pollination. See "Small-scale beekeeping".
5. **Pigs, sheep and cattle**: need more land and feed; sheep give wool (see "Textiles and leather"), cattle milk and traction.

Animal health — signs of illness

Isolate any animal that: stops eating, is dull, has diarrhoea, cough, nasal/eye discharge, lameness, weight loss or fever. Wash hands before and after. New animals stay in quarantine 2–3 weeks before joining the group.

- **Parasites**: rotate pastures, clean bedding, check the inner eyelid colour (pale = worm anaemia; see the FAMACHA method in the manuals).
- **Wounds**: clean, remove maggots, disinfect, protect from flies.
- **Birth**: mostly unaided; intervene only if straining >30–60 min without progress — clean, lubricated hands, pull with contractions, down and back.
- **Zoonoses** (spread to people): rabies (bites — wash 15 min), brucellosis (boil milk!), tuberculosis, salmonella (wash dirty eggs, cook meat well), mange, ringworm.

Feeding

Pasture and hay (cut and dry grass in summer for winter), garden scraps, cracked grain, acorns (pigs), dried nettles (hens). Clean water always. Salt/minerals for ruminants.

Using the products

- **Milk**: milk with clean hands and udder; **boil or pasteurise** (72 °C for 15 s or 63 °C for 30 min). Cheese: rennet (kid/calf stomach) or acid (lemon, vinegar); see "The Book of Cheese".
- **Slaughter and meat**: stun first (firm blow to the head), bleed at once, hang, skin/pluck, gut without tearing intestines, cool. Preserve: salt, smoke, dry, lard.

- **Hides and wool**: see "Textiles and leather". **Bones**: tools, needles, glue (boil). **Fat**: candles, soap ("Basic chemistry"). **Manure**: compost and, dried, fuel.

Animal traction

Donkeys, horses and oxen: well-fitted yoke/collar, trimmed hooves, rest and water every 1–2 h of work. Plough, cart, water wheel (noria), millstone.

17. Natural building — durable shelter, stoves, wells, latrines and water

Oracle OS guide. Manuals in the library: "Shelters, Shacks and Shanties", "The Useful Arts Employed in the Construction of Dwelling Houses", "The Science of Brickmaking", "Concrete Construction for the Home and the Farm", "Practical Masonry", Practical Action (rainwater, pumps), WEDC/WHO (sanitation), Appropedia, Open Source Ecology.

Choosing a site

Slightly raised ground (no flooding), near water but latrines more than 30 m from it, sheltered from the prevailing wind, winter sun (south-facing in the northern hemisphere), firewood and good soil nearby.

Wall techniques (simplest to most durable)

- **Log-and-branch hut**: A-frame covered with boughs/foliage/turf — days.
- **Wattle and daub**: stakes and woven rods (willow, hazel) daubed both sides with clay + straw + dung.
- **Cob**: clay + sand + straw trodden together, stacked in 30–50 cm lifts, walls 40–60 cm thick. Great thermal mass.
- **Adobe**: clay + straw blocks moulded and sun-dried 2–4 weeks.
- **Rammed earth**: damp soil tamped between timber forms in 10–15 cm layers.
- **Stone** with lime mortar (see "Basic chemistry"), the most durable; walls wide at the base.
- **Timber frame**: posts and beams with mortise-and-tenon joints (see "Smithing, pottery and carpentry").
- **Golden rule for earth walls**: "**good boots and a hat**" — a stone base above splash height (30–50 cm) and wide eaves so rain never wets the wall.

Roof

Pitch ≥ 45° for thatch (rye/wheat straw, reed) 30 cm thick; fired tiles; slate; overlapping boards; turf over birch bark. Ventilate the roof space.

Insulation and floor

Straw, wool, dry bracken, cork in wall cavities; rammed earth floor over a stone/gravel layer (stops damp) finished with clay + linseed oil.

Stove, chimney and heat

- **Rocket stove**: insulated "L" tube (brick, clay + ash) — wood feeds from the side, flame rises in an insulated chamber; clean burn with little wood.
- **Masonry/thermal-mass stove**: hot smoke runs through a long channel inside a clay/stone bench that stores heat for hours.
- **Chimney** 1 m above the highest point of the roof; sweep once or twice a year (soot fires). Never an indoor fire without a chimney (carbon monoxide).
- **Bread oven** of clay (dome on a stone-and-sand base).

Water

- **Hand-dug well**: only by trained people, with shoring (brick/concrete/timber rings) — collapses and gases kill. Line the top, lid, raised rim, no latrine within 30 m or uphill.
- **Rainwater harvesting**: clean roof → gutter → first-flush diverter → dark covered tank → tap. 1 mm of rain on 1 m² = 1 litre.
- **Slow sand filter** (see Practical Action): 60–90 cm of fine sand over gravel; the top biological layer cleans the water; then disinfect.

Sanitation

- **Pit latrine**: 2–3 m pit, slab with lid, shelter; 30 m from wells and above the water table; ash or lime after each use. When full to 50 cm from the top, cover with soil and dig another.
- **Composting dry toilet**: faeces + sawdust/straw, compost 1–2 years before using on trees (never on raw-eaten vegetables).
- Handwashing next to the latrine (tippy-tap: hanging jerrycan with a rope foot pedal).

18. Off-grid energy — charging Oracle and working without mains power

Oracle OS guide. Manuals in the library: "Electricity for the Farm" (water wheel and dynamo), "Windmills", Energypedia, Appropedia, Practical Action (solar PV, hand pumps).

1. Priority: keep Oracle alive

- **Consumption**: a phone uses ~10–20 Wh per full charge. A 20 W USB solar panel in direct sun gives ~60–80 Wh per day → several charges.
- **USB solar panel**: face the sun (south in the northern hemisphere), tilt ~ your latitude (Scotland ~55°, Portugal ~40°), no shade; re-aim 2–3 times a day. On cloudy days it gives 10–25%.
- **Power bank**: charge the power bank by day and the phone at night from it (protects the phone from fluctuations). LiFePO4 lasts 2,000–4,000 cycles; ordinary lithium 300–500.
- **Batteries**: store at 40–80% charge; avoid heat (never in sun behind glass) and extreme cold while charging (below 0 °C causes damage).
- **Save**: airplane mode, minimum brightness, Oracle's extreme power-saving mode, local AI only when needed (it drains a lot), switch the phone off when not in use.
- **Redundancy**: a spare phone switched off in a metal bag/closed tin (Faraday), charged to 50% and checked every 6 months.

2. Making electricity without a panel

- **Bicycle dynamo**: a hub dynamo (6 V, 3 W) with a dynamo USB charger charges a phone in 2–4 h of pedalling. The bike can be mounted on a fixed stand.
- **Car alternator / DC motor**: turned by crank, water wheel or wind, charges a 12 V battery; then a 12 V→USB converter (car socket).
- **Car batteries**: store energy; don't discharge below ~50% (12.2 V); lead-acid gives off hydrogen — charge outdoors.
- **Charge controller**: always between generator/panel and battery, or it will be destroyed.

3. Mechanical power (what moves a society)

- **Human power**: cranks, pedals, treadle loom and lathe, hand pumps (see Practical Action).
- **Animal traction**: horses, donkeys, oxen — ploughing, hauling, turning mills (horse gin).
- **Water wheel**: needs a drop or steady current. An overshot wheel is most efficient with little water and a 2–10 m drop; undershot for a wide river with little drop. It grinds grain, saws, pumps, powers trip hammers and generates electricity.
- **Windmill**: canvas or wooden sails on a frame facing the prevailing wind; grinding and pumping water.

- **Levers, pulleys, ramps, wheels** — see "Ropes and knots" (mechanical advantage).

4. Heat and light

- **Firewood**: season 1–2 years; green wood gives little heat and lots of smoke. Oak, ash and beech are best.
- **Charcoal**: burns hotter and cleaner (forge, cooking) — see "Basic chemistry".
- **Rocket stove and hay box** (retained-heat cooking) save half the wood — see "Natural building" and "Heat and cook with no power".
- **Light**: tallow or wax candles, vegetable-oil lamps with cotton wicks, dynamo torches; rise and sleep with the sun.
- **Solar hot water**: black hose or a drum painted black in the sun.

5. Biogas (communities, year 2+)

Manure + water in a sealed digester makes methane for cooking; needs careful building (leaks are dangerous). See Energypedia/Appropedia.

19. Basic chemistry for rebuilding — soap, lye, lime, charcoal, candles, vinegar, alcohol, ink and paper

Oracle OS guide. Manuals in the library: "Soap-Making Manual", "Handbook of Soap Manufacture", "Vegetable Dyes", LibreTexts Chemistry, Wikipedia. Wear gloves and eye protection and work outdoors: lye and lime burn skin and eyes (rinse 15 min with water if it happens).

Wood-ash lye (potash) — the base of soap

1. **White** ash from hardwood (not paper, plastic or barbecue briquettes) in a bucket with holes in the bottom lined with straw.
2. Pour rainwater slowly; collect the brown liquid. Run it through the ash again until strong: **an egg or potato floats** showing a coin-sized patch above the surface.
3. Boil down to concentrate if needed. Store in glass or clay (not aluminium).

Soap

- **Fat**: beef/mutton tallow, lard, olive or other oils; melt and strain.
- **Soft soap** (with ash lye = potassium): boil fat and lye, stirring for hours, until a honey-like paste with no fat on top. Test: a cooled dab should not strongly sting the tongue or feel greasy.
- **Hard soap**: add salt at the end (the paste separates and soap rises), or use bought caustic soda (weigh with a saponification table).
- Cure 4–6 weeks. Washing hands with soap prevents most diarrhoea and infections.

Lime (mortar, limewash, disinfection, nixtamal)

- Burn limestone (or shells) in a kiln at ~900 °C for 1–3 days → quicklime. **Very caustic and heats with water**: slake slowly outdoors → slaked lime (putty). Store covered with water.
- Mortar: 1 part lime putty + 3 parts sand. Limewash: lime + water like milk — disinfects and protects walls. Lime in latrines cuts flies and smell.

Charcoal

Stack dry wood, cover with earth/turf leaving vents, light from the centre; it smoulders 1–3 days with little air; when smoke turns thin and blue, seal everything and let it cool 1–2 days. Used in forges, water filters (crushed charcoal between sand) and smokeless cooking.

Candles and light

Melted tallow or beeswax; dip a cotton wick repeatedly (dipped candles) or pour into a mould. Oil lamp: olive/vegetable oil in a dish with a wick. Never leave a flame unattended.

Vinegar and preserving

Wine, cider or diluted honey fermented + vinegar "mother", in an open vessel covered with cloth, 3–6 weeks somewhere warm. Pickles vegetables and cleans.

Alcohol (antiseptic and fuel)

- Ferment fruit, honey or malted grain with yeast (or sourdough) for 1–2 weeks.
- **Distil** in a still (closed pot + copper/glass tube cooled in water): **discard the first 5% (contains methanol, toxic)**. Disinfecting skin and instruments needs 60–80% alcohol; so does burning in lamps.
- Caution: flammable vapours — keep away from open flames.

Salt

Evaporate seawater in trays in the sun or by boiling; strain through cloth first. Essential for preserving meat/fish and for livestock.

Ink, paper and glue

- **Ink**: finely ground soot/charcoal + gum arabic or glue + water; or iron-gall ink (oak galls + iron sulphate made from scrap iron in vinegar).
- **Paper**: cotton/linen rag fibres or plants (nettle, straw) boiled with ash, beaten to pulp, spread in water and lifted on a mesh frame; press and dry.
- **Glue**: boil hides, bones or hooves for hours (animal glue); cooked starch (flour paste).

Dyes

See "Textiles and leather" and "Vegetable Dyes": onion skins (yellow), walnut (brown), madder (red), indigo/woad (blue); alum or iron mordant.

20. Smithing, pottery and carpentry — making and repairing tools, vessels and structures

Oracle OS guide. Manuals in the library: "Forge Work" (Ilgen), "Practical Forging and Art Smithing", "The Potter's Craft", "Pottery for Artists", "How to Make Pottery", "Woodwork Joints", "Carpentry and Woodwork", "Rustic Carpentry", "Things To Make"; Gutenberg technology (illustrated, in Encyclopedias); Open Source Ecology.

Smithing (start with scrap)

- **Raw material**: car leaf springs, old files, rebar, rails, plate — there is salvageable metal for decades.
- **Forge**: **charcoal** fire in a pit or a stone/brick box with air blown from below (hide bellows, old hairdryer, crank blower). Anvil: heavy steel block, rail section or flat hard stone. Tongs, 1–1.5 kg hammer, bucket of water.
- **Steel colours** (in shade): dull red ~700 °C, orange ~900 °C (working), yellow ~1,100 °C (forge welding), white with sparks = burning (ruined).
- **Operations**: drawing out (lengthening), upsetting (shortening and thickening), bending, hot punching, cutting with a chisel, forge welding (borax or fine sand as flux).
- **Hardening** (carbon steel, e.g. springs, files): heat until a magnet no longer sticks, quench in oil/water → hard and brittle; then **temper**: clean and heat slowly to straw colour (knives, chisels) or blue (springs), cool.
- First useful items: nails, hooks, hinges, knives, chisels, hoe blades, horseshoes, forge tools.

Pottery

- **Clay**: sticky soil from riverbanks/ditch bottoms; dry, break up, soak, strain through cloth, drain to plasticine consistency; mix sand/ground fired clay (grog) to stop cracking.
- **Forming**: coils stacked and smoothed; slabs; moulding in a basket; kick wheel if possible.
- **Dry** slowly in shade 1–2 weeks (cracks = dried too fast).
- **Fire**: pit fire (600–800 °C, porous ware) or brick/clay kiln (900–1,100 °C, stronger); heat slowly over hours.
- **Waterproofing**: burnish before firing, glaze (ash + clay + sand in a hot kiln), or seal with milk/fat.
- **Bricks**: clay + straw/sand in wooden moulds, sun-dried (adobe) or clamp-fired (brick).

Hand-tool carpentry

- **Basic tools**: axe, crosscut and rip saws, chisels, plane, brace and bit, mallet, square, plumb line, water level (clear hose with water), file, sharpening stone.
- **Sharpening**: 25–30° angle, wet stone, steady strokes until a burr forms; remove it on the other side.

- **Timber**: fell in winter, season 1 year per 2.5 cm thickness, stacked with spacers, in shade.
- **Joints** (no nails): half-lap, mortise and tenon (strongest for frames), dovetail (drawers and boxes), wooden pegs.
- **Priority items**: tool handles (ash), gates, benches and tables, beds, window frames, beehives, wheelbarrows, wheels, barrels.
- **Green wood**: carve spoons and bowls, peg-jointed chairs (easy to work fresh).

Safety

Smithing: goggles, leather apron, closed shoes, water nearby, never quench indoors. Pottery/kilns: fire outdoors, watch smoke and burns. Carpentry: always cut away from the body; sharp tools are safer than blunt ones.

21. Textiles and leather — fibres, spinning, weaving, dyeing, sewing, tanning and shoemaking

Oracle OS guide. Manuals in the library: "Hand-Loom Weaving", "Textiles" (Dooley), "Textiles and Clothing", "Vegetable Dyes", "A Text-book of Tanning", "Leather" (Adcock); Gutenberg technology (illustrated).

Fibres

- **Wool** (sheep, goat, angora rabbit): shear in spring, wash in warm soapy water without rubbing (it felts), dry.
- **Flax**: sow in spring, pull when yellow, **ret** (rot the stems in water or dew 2–4 weeks), dry, break and scutch to free the fibre, hackle (comb).
- **Nettle and hemp**: like flax; strong fibre for rope and sacks.
- **Reuse**: unravel old fabric and knitwear — the yarn already exists.

Carding and spinning

- Hand cards (nail brushes) align fibres into rolls.
- **Drop spindle** (stick with a weight): attach fibre, spin the spindle and let twist climb while drawing the fibre out slowly; wind on and repeat. A spinning wheel is faster.
- **Plying** two yarns in the opposite direction gives stronger, balanced yarn.

Weaving, knitting and sewing

- **Simple loom** (frame or rigid heddle): stretch the warp (vertical threads) and pass the weft over and under, beating it tight. See "Hand-Loom Weaving".
- **Knitting and crochet** with wooden or bone needles; socks, hats, jumpers.
- **Sewing**: backstitch (strong), running stitch, patches; needles from bone or sharpened, drilled wire.
- **Felt**: wool + hot water + soap + friction → dense fabric for boots, hats and blankets.

Dyeing

Mordant first (alum, or an iron pot), then a hot dye bath for 1 h: onion skins (yellow/orange), walnut hulls and leaves (brown), madder (red), woad/indigo (blue), heather and broom (yellow), iron (darkens everything).

Tanning hides

1. **Skin** carefully; **flesh** (scrape fat and meat) over a log.
2. **Hair off** (leather): soak in water with ash or lime 3–7 days until hair slips; scrape.
3. **Bark tanning** (tannin): ground oak, chestnut or willow bark in water; hide in it 4 weeks to several months (thick leather), strengthening the solution gradually. Gives hard, lasting leather (shoes, belts, harness).
4. **Brain/oil tanning** (soft hide, buckskin): spread a paste of cooked brain or egg yolk, work the hide by stretching as it dries, **smoke** it so it doesn't stiffen in rain.

5 Dress with tallow or oil.

Shoes

Thick leather sole (or old tyre), soft leather upper, sewn with waxed linen thread or leather thongs; pattern made from the foot. Tyre-and-strap sandals are quickest. Dry socks and footwear are essential for walking and working (see "Stretching and pain" — feet).

22. Ropes, knots and lashings — essentials for shelter, loads, rescue and fishing

Oracle OS guide. Based on US Army FM 3-97.61 (Military Mountaineering), the FAO Fisherman's Workbook and A. Hyatt Verrill's "Knots, Splices and Rope Work" (in the library **with step-by-step illustrations**; also the "Knots and ropes (zimgit)" collection in the encyclopedia reader). To hang people on a rope (climbing, lowering, rescue at height) use certified gear and training: one mistake kills.

Know your rope

- **Synthetic** (nylon, polyester, polypropylene): strong, rot-proof. Nylon stretches (absorbs shock); polyester stretches little (lashings); polypropylene floats (water rescue) but degrades in sunlight.
- **Natural** (sisal, hemp, cotton): weaker, rot when damp, but hold knots well.
- **Static** (low stretch) for rescue and loads; **dynamic** (stretchy) only for lead climbing.
- A knot cuts rope strength by **20–50%**. Load only a fraction of breaking strength: **1/5 for loads**, far less for people.
- Care: keep away from sand, chemicals, battery acid, sharp edges (pad them), prolonged sun. Dry in shade. Retire rope with stiff, soft, fuzzy or melted sections.
- Ends: melt synthetics or **whip** them with thread so they don't fray.

Terms

Working end, **standing part**, **bight** (U-shaped fold), **loop** (crossed). Always **dress** (tidy and tighten) the knot and leave a tail of at least 10 rope diameters.

The most useful knots

1. **Figure-eight stopper**: make a bight, take the working end round the standing part and back through the bight — it forms an "8". Stops the rope pulling through.
2. **Figure-eight follow-through** (strong fixed loop): tie a loose figure-eight 1 m from the end, pass the end around the anchor (or harness) and **retrace the eight exactly**. Standard tie-in knot; easy to check and untie.
3. **Bowline** (non-slip loop): small loop in the standing part ("the hole"), the end ("the rabbit") comes up through the hole, round the tree (standing part) and back down the hole. Dress it. Can loosen without load: back it up with a stopper.
4. **Clove hitch**: two overlapping turns round a post, the second tucked under the first. Quick start for lashings and tying to posts; adjustable.
5. **Round turn and two half hitches**: two full turns round the post, then two half hitches on the standing part. Holds under load and releases easily. Great for boats and tarps.
6. **Taut-line hitch**: round the peg, two turns on the standing part inside the loop and one outside. Slides to tension a guy line and grips under load.
7. **Trucker's hitch**: make a fixed loop in the standing part, pass the end round the hook/peg, back through the loop and pull — **3:1 advantage**. Finish with two half hitches. For loads, clotheslines and shelter ridgelines.

8. **Sheet bend**: joins ropes of different thickness. Bight in the thick rope; thin rope comes up through the bight, round both legs and under itself. **Double sheet bend** for more security.
9. **Double fisherman's**: joins two equal ropes under load (closes cord loops). Each end ties a double overhand (two turns) round the other; they jam together.
10. **Prusik**: loop of thinner cord (5–7 mm on 10–11 mm rope) wraps the main rope 3 times and passes through itself. Grips when loaded, slides when not: ascending, backing up a lower, rescue systems.
11. **Munter hitch**: on a carabiner, creates friction to hold or lower a load.
12. **Timber hitch**: round a log, the end wraps round itself 3 times. For dragging or lifting logs; add a half hitch ahead to steer.
13. **Reef knot**: right over left, then left over right. **Bandages and parcels only** — slips under load, never to join load-bearing ropes.
14. **Improved clinch** (fishing): line through the hook eye, 5–7 turns round the standing line, back through the first loop by the eye, then through the big loop; wet and pull tight.

Lashings for building (shelters, tripods, stretchers, rafts)

- **Square lashing** (crossed poles): clove hitch on one pole, 3–4 wraps around both, 2–3 **frapping** turns between the poles, finish with a clove hitch.
- **Diagonal lashing**: when poles want to spring apart — diagonal wraps one way, then the other, frapping, clove hitch.
- **Shear / tripod lashing**: poles side by side, 6–8 loose wraps, 2 frapping turns between them; open into an A-frame or tripod (hang a pot, support a shelter, make a trestle).
- **Shelter ridgeline**: rope between two trees with a round turn at one end and a trucker's hitch at the other; tarp over, corners tied with taut-line hitches.

Mechanical advantage (moving weight)

- A fixed pulley only changes direction. A **Z-drag (3:1)**: rope tied to the load runs through a pulley/carabiner at the anchor, back to a second pulley clipped with a prusik onto the rope near the load, then to the pullers. One third of the effort (minus friction).
- Anchors: thick living trees, large boulders, vehicles (braked). Share the load across more than one anchor.

Making cordage without rope

- Fibres: dried nettles, inner bark of lime/willow/oak, cattail leaves, palm raffia, plastic bags cut into strips, cloth strips.
- **Reverse wrap**: fold a bundle of fibres in half; twist the top strand away from you and cross it over the other toward you; repeat alternately. The cord locks itself. Splice in new fibres before a strand runs out.
- A three-strand braid works too but is weaker.

Practise

Practise each knot until you can tie it eyes closed and with gloves. Priority: figure-eight follow-through, bowline, round turn and two half hitches, trucker's hitch, sheet bend, prusik.

23. Navigation without internet — map, compass, sun, stars and sea

Oracle OS guide. Based on US Army FM/TC 3-25.26 (Map Reading and Land Navigation), Ordnance Survey "Map reading made easy" and Bowditch "American Practical Navigator" (in the library **with illustrations**). Oracle shows your GPS position and coordinates without signal (Map tab) and Organic Maps routes offline; this guide is for when the phone fails.

Reading the map

- **Scale**: 1:25 000 → 4 cm = 1 km; 1:50 000 → 2 cm = 1 km.
- **Contour lines** join equal heights. Close together = steep; far apart = gentle. "V"s pointing uphill = valley/stream; pointing downhill = spur/ridge.
- **Grid reference** (OS maps in the UK, also UTM/MGRS): numbers along the bottom first (**"along the corridor, then up the stairs"**), then the side. 6 figures = 100 m precision.
- **Set the map**: turn it until map features line up with the ground (or map north with compass north).

Compass

- **Take a bearing from the map**: lay the compass edge between where you are and where you're going (direction arrow toward the target); turn the housing until its lines are parallel to the map's north–south lines, N to north; read the bearing.
- **Magnetic declination**: difference between grid north and magnetic north. In the UK and Portugal it is small (near 0°) this decade, but check the map margin — elsewhere it can be 10–20°.
- **Walk a bearing**: hold the compass in front, turn your body until the red needle sits in the housing arrow; pick a landmark in that direction, walk to it, repeat.
- **Back bearing** (return): add or subtract 180°.
- **Fix your position (resection)**: take bearings to two or three known visible features (peaks, masts), convert to back bearings, draw the lines on the map; they cross where you are.
- Keep the compass away from metal, magnets, phones and cars.

Techniques good navigators use

- **Handrails**: follow linear features (river, wall, road, power line).
- **Catching feature**: something that tells you you've overshot (a river, a road).
- **Aiming off**: to reach a bridge on a river, deliberately aim to one side; when you hit the river you know which way to turn.
- **Pacing**: count each time the same foot lands; an adult takes ~60–70 double paces per 100 m on the flat (calibrate yours). More uphill, in rough ground or at night.
- **Timing**: walking ~4–5 km/h on the flat + 1 minute per 10 m of climb (Naismith's rule).

Without a compass

- **Sun**: rises roughly east and sets roughly west (further north-east/north-west in summer). At solar noon it is due south (northern hemisphere) or north (southern hemisphere).
- **Shadow stick**: plant a vertical stick; mark the shadow tip; wait 15–20 min and mark again. The line from first to second mark runs **west to east**.
- **Analogue watch** (northern hemisphere): point the hour hand at the sun; south is halfway between it and 12 (use 1 o'clock during summer time). Southern hemisphere: point 12 at the sun; north is halfway between 12 and the hour hand.
- **Pole Star** (northern hemisphere): in the Plough (Big Dipper) extend the line through the two end "pointer" stars ~5 times; it reaches Polaris, which marks north and doesn't move.
- **Southern Cross** (southern hemisphere): extend the long axis ~4.5 times; drop vertically to the horizon: south.
- Moss, anthills and "the side of trees" are **not reliable**.

Coordinates and GPS

- Formats: decimal degrees (55.8642, -4.2518), degrees-minutes-seconds, UTM/MGRS, OS grid. Always state the format when passing coordinates by radio.
- Write down coordinates of shelter, water and meeting points on paper.
- Phone GPS works without signal; save battery with airplane mode and location on.

At sea and on rivers

- **Nautical charts**: depths in metres (soundings), hazards, lights with their character (e.g. Fl(2) 10s = 2 flashes every 10 s).
- **IALA A buoyage** (Europe, Africa, most of Asia and Oceania): **entering** harbour, **red to port** (left), green to starboard. **IALA B** (Americas, Japan, Korea, Philippines): the reverse ("red right returning").
- **Cardinal marks** (yellow and black, two cones): pass on the named side. North: cones up, black over yellow; South: cones down, yellow over black; East: cones base to base, black-yellow-black; West: cones point to point, yellow-black-yellow.
- **Tides — rule of twelfths**: over ~6 hours the water rises/falls 1/12, 2/12, 3/12, 3/12, 2/12, 1/12 of the range each hour. Streams are strongest in the middle hours.
- **Dead reckoning**: distance = speed × time; log course, time and speed every hour on paper.
- **Collision rules**: overtaking vessel keeps clear; head-on both turn to starboard (right); crossing — give way to the vessel on your right; power gives way to sail (except in narrow channels). Lights: red port, green starboard, white stern and masthead.
- Rivers: run the outside of bends (deeper) but watch for fallen trees; never approach weirs and dams from upstream.

Plan before you go

Tell someone your route, return time and who is going. Carry a paper map, compass, whistle, torch, spare clothing, water and food.

24. Water, rivers, lakes and sea — currents, crossings, floods and survival in water

Oracle OS guide. Based on RNLI ("Float to Live"), NOAA (rip currents), US Coast Guard (Boat Crew Seamanship, Rescue and Survival), SOBRASA (aquatic emergencies) and US Army FM 3-05.70 (crossings) — in the library **with illustrations**. For making water safe to drink see "Drinking water".

Cold water shock — "float to live"

- Falling into water below 15 °C causes **gasping and uncontrolled breathing** in the first minute: that is when many people drown.
- **Don't swim straight away. Lean back, tilt your head back, spread arms and legs, and float for 60–90 seconds** until your breathing is under control. Then swim to safety or call for help.
- With a life jacket, waiting for rescue: **HELP position** (knees to chest, arms crossed) or as a group **huddle in a circle** to keep heat. Keep clothes on (insulation).
- After pulling someone from cold water, treat hypothermia ("First aid" guide); keep them lying down and watched — they can deteriorate.

Rip currents at the beach

- Spot them: a **darker, calmer** channel between breaking waves, foam or debris moving out to sea, choppy brown water.
- If caught: **don't swim against it**. Float, raise an arm and shout. Swim **parallel to the shore** until out of it (usually 10–30 m wide), then angle back in with the waves.
- To help someone: don't go in. Throw something that floats, call 999/112 and ask for the Coastguard (UK).

Tides, beaches and coast

- Check tide tables; on wide sands and estuaries the tide can come in faster than you can walk. Always know each beach's exit.
- Mud and quicksand: don't struggle; lean back, spread your weight, move slowly backwards. Call 999/112.
- Bigger waves come without warning: keep back from rock and pier edges, never turn your back on the sea.

Rivers and moving water

- **15 cm of fast water can knock an adult over; 30–60 cm can float a car.** In floods: **turn around, don't drown**. Never drive through flooded roads.
- If you fall into a fast river: **defensive swimming position** — on your back, feet first and at the surface (pointing downstream), use your feet to fend off rocks, arms to steer. **Don't try to stand** in fast water above the knees: your foot can wedge between rocks and the current pushes you under.
- **Strainers** (fallen trees, grilles, debris on piers): water passes through, people don't. If heading into one, turn onto your front and swim hard **over** it.

- **Weirs and low dams**: the recirculating "hydraulic" below traps people and boats. Never approach from below or above. To escape: dive deep and exit underneath, with the current.

Crossing a river (only if there is no alternative)

1. Look for a bridge, known ford, or the **widest, braided** section (shallower and slower). Avoid outside bends and just above rapids, drops or obstacles.
2. Check depth and force: if it is over the knee and fast, **don't cross**.
3. Unclip pack hip belt and chest strap (so you can ditch it), keep shoes on.
4. Use a **strong pole as a third leg** on the upstream side; move sideways facing upstream, short steps, one point of contact moving at a time.
5. In a group: **in line with the current** (strongest upstream, others holding the waist of the one in front) or **huddle/triangle** with arms linked.
6. Never tie someone to a rope for a crossing: a person on a tight rope in current gets pulled under.

Lakes and ice

- Safe ice to walk on: **at least 10 cm of clear, hard ice**. White, grey, snow-covered ice or ice near stream inlets is weaker.
- If you fall through: turn toward where you came from (that ice held you), arms on the ice, **kick like swimming** and pull yourself out horizontally; **roll** away from the hole. Don't stand until on safe ice.

Survival at sea

- **Stay with the boat** (even capsized): bigger and easier to spot. Wear a life jacket, warm and waterproof clothing.
- **Never drink seawater or urine.** Ration fresh water (without water, don't eat — digestion uses water). Catch rain on tarps; solar still in a bucket.
- Shelter from sun and wind; stay dry; seasickness dehydrates — lie down and look at the horizon.
- Signals: VHF channel 16 ("MAYDAY" ×3, vessel name, position, problem, people aboard), flares, mirror, whistle, bright clothing. EPIRB/PLB if you have one.
- Only abandon ship to a liferaft **when the boat is really sinking** ("step up into the raft", don't jump into the water).

Drowning — recognise and rescue

- Drowning people **rarely shout or wave**: upright, head back, mouth at water level, not moving forward. Ask "Are you OK?" — no answer, act.
- **Reach, throw, row, go**: reach with a pole/rope/towel while lying down; throw something that floats; use a boat; only go in trained and with buoyancy.
- Out of the water: unresponsive and not breathing normally → **5 rescue breaths then CPR 30:2** (see "First aid"). Even if they recover, they must see a doctor.

Small boats and rafts

- Weight low and central, three points of contact when boarding, always a life jacket. Don't overload.
- **Emergency raft**: sealed empty drums/jerrycans or dry logs, fixed to a pole frame with square lashings ("Ropes and knots"). Test in calm shallow water.

Lakes and rivers as a resource

- Surface water must be **treated** before drinking (boil 1 minute). Collect upstream of settlements, animals and latrines; never drink water with odd foam, chemical smell or blue-green algae (cyanobacteria — boiling doesn't destroy their toxins).
- Wash yourself and clothes **60 m from the water**, using a container, so the source stays clean.

25. Communication without power — signals, Morse code, runners and simple radio

Oracle OS guide. Manuals in the library: "Boy Scouts Handbook" (signals, semaphore, Morse), FM 21-76 (distress signals), ARRL/amateur radio (if available), Appropedia (crystal radio).

International distress signals

- **Threes of anything**: 3 fires in a triangle, 3 whistle blasts, 3 shots, 3 flashes — repeat every minute.
- **SOS** in Morse: · · · — — — · · · (light, mirror, whistle, knocking).
- **Ground-to-air signals** (letters 3 m tall, high contrast — stones, branches, clothing, trenches in snow): **V** = need help; **X** = need medical help; **N** = no; **Y** = yes; **→** = going this way.
- **Signal mirror**: stretch your arm out, make a "V" with your fingers around the target and steer the reflection into the V. Visible tens of km away in sunshine.
- Smoke: by day, white smoke (green leaves, moss) against dark backgrounds; black smoke (rubber, oil) against snow. At night, flame.

Morse code (learn A–Z and 0–9)

A ·— B —··· C —·—· D —·· E · F ··—· G ——· H ···· I ·· J ·——— K —·— L ·—·· M —— N —· O ——— P ·——· Q ——·— R ·—· S ··· T — U ··— V ···— W ·—— X —··— Y —·—— Z ——·· 1 ·———— 2 ··——— 3 ···—— 4 ····— 5 ····· 6 —···· 7 ——··· 8 ———·· 9 ————· 0 ————— Timing: dot = 1, dash = 3, gap within a letter = 1, between letters = 3, between words = 7. Means: shuttered lamp/torch, mirror, flag (right = dot, left = dash), whistle, drum, knocking on pipes.

Community codes and messages

- **Bells/drums/horns** with agreed patterns: 1 long = meeting; rapid repeat = fire/danger; 3 = medical help. Post the table in a public place.
- **Flags/cloths** on visible high points (e.g. white = all well, red = danger, yellow = sickness/quarantine).
- **Central message board** in each settlement: date, from, to, message. Write with charcoal or chalk.
- **Runners**: on foot ~5 km/h, on horseback 10–15 km/h; written message, folded and marked; agreed route and relay points (like the old posts). Two copies by different routes for vital messages.
- **Homing pigeons**: return to their home loft (up to hundreds of km); breed and train from lofts in each settlement.
- **Heliograph and semaphore**: towers in line of sight every 5–15 km can pass a message across a region in a day.

Radio without the grid

- A **hand-crank/solar radio** (AM/FM/shortwave) keeps you informed — store it in a metal box (protection from electromagnetic pulses) with rechargeable batteries.
- **Crystal radio**: receives AM with no batteries — long antenna (20–30 m of wire up high), ground connection (pipe/damp stake), coil (wire wound on a tube), diode (galena, pyrite or a salvaged germanium diode), high-impedance earphone. See Appropedia/Boy Scouts.
- **PMR446/CB/VHF** radios with a solar panel and rechargeable batteries: 1–5 km range (PMR), more with high antennas. Agree channels and listening times (e.g. every hour on the hour, 5 min).
- **Amateur radio (HF)**: continental range; needs a licence in normal times and know-how — train operators now.

Durable writing and records

- **Oak-gall + iron sulfate** ink, or soot + gum arabic; charcoal; cut goose quills.
- Handmade paper from cotton/linen rags or plant fibres (see Textiles), parchment from hide.
- Copy Oracle's vital documents onto paper (Print pack) and store them in more than one dry place.

26. Communication without internet: AM, FM, long wave and two-way radio (English)

Oracle OS guide. See also the Oracle **Radio** screen (official stations per country).

Listening — no licence needed

- **FM (87.5–108 MHz)**: tens of km. Phones with FM need **wired headphones** (antenna); works without SIM, internet or in airplane mode.
- **AM / medium wave**: hundreds of km, especially at night; needs an AM radio (most phones lack it).
- **Shortwave**: international news.
- **Long wave**: BBC Radio 4 on 198 kHz was switched off on 27 June 2026.
- A **wind-up/solar AM/FM/SW radio** is the most reliable kit item. Save batteries: listen **on the hour** for news.
- In a UK national emergency the BBC (Radio 4 and nations/local stations such as BBC Radio Scotland) carries government announcements.

Talking — two-way radio

- **PMR446 (Europe/UK)**: licence-free walkie-talkies, 446 MHz, 0.5 W, 16 channels, typically 1–3 km in towns. Agree a **channel and schedule** with family (e.g. channel 3, 5 minutes every hour on the hour).
- **CB 27 MHz**: licence-free in UK/EU (CEPT); **channel 9** commonly used for emergencies, 19 for road traffic.
- **USA**: FRS (licence-free), GMRS (simple licence), CB channel 9.
- **Amateur radio**: **licence required**; amateurs run emergency nets (ARES, RAYNET) — see ARRL ARES manual in the library. In imminent danger to life, ITU Radio Regulations allow any means to call for help.
- **Marine VHF channel 16 (156.8 MHz)**: distress at sea.

Calling for help

"**MAYDAY ×3**" (danger to life) or "**PAN-PAN**" (urgent). Who you are, **where** (coordinates from Oracle's Emergency screen), what happened, how many people, what help. Speak slowly, repeat, **listen**.

Signals without radio

Three of anything = distress (3 whistle blasts, lights, fires). Alpine: 6 blasts per minute. Morse SOS ··· ––– ···. Ground-to-air (3 m letters): **V** need help, **X** need medical help, **N** no, **Y** yes, **→** going this way. Signal mirror or phone screen to reflect sunlight.

Mobile phone

112 works across Europe and in many countries on any available network; many phones call 112 without a SIM. SMS gets through when calls fail. Save battery: airplane mode between checks. Keep cell-broadcast emergency alerts enabled.

27. Weather and calendar — forecasting, telling time, planning the year

Oracle OS guide. Manuals in the library: "Weather Proverbs and Paradoxes", "The Weather and Practical Methods of Forecasting", "Weather Lore", "Clouds and Weather" (Gutenberg), Boy Scouts Handbook, navigation (stars).

Reading clouds

- **Cirrus** (high wisps) thickening into **cirrostratus** (veil, halo round sun/moon) → rain in 12–24 h (warm front).
- Grey **altostratus** lowering → steady rain on the way.
- Fair-weather **cumulus** (cotton, flat base) in the morning = fine; growing into towers in the afternoon (**cumulonimbus**, anvil top) = thunder, hail, gusts.
- Low **stratus**/fog in the morning that lifts = good day.
- "Mackerel sky" (**altocumulus** scales) on a summer morning = afternoon thunderstorms.

Other signs

- Wind veering clockwise (northern hemisphere) and strengthening → front passing; moist south/south-westerly wind → rain in western Europe.
- "Red sky at night, shepherd's delight; red sky in the morning, shepherd's warning" (works with weather arriving from the west).
- Morning dew/frost = clear night, no rain imminent. No dew on a clear night = wind or cloud coming.
- Smoke sinking and spreading, sounds carrying, strong smells, birds flying low, insects biting → falling pressure, rain.
- **Thunderstorms**: seconds between flash and thunder ÷ 3 = km. Under 30 s, take shelter (not under lone trees; crouch with feet together if exposed).
- **Improvised barometer**: jar of water with an inverted straw/tube, or a balloon stretched over a jar with a pointer — the trend matters more than the value: falling fast = bad weather.
- **Frost**: clear, still, dry nights — protect plants (cover, water in the late afternoon).

Telling the time

- **Sundial**: vertical stake, mark the shadow hourly for a day; shortest shadow = solar noon. For accuracy: gnomon tilted at an angle = latitude, pointing at the pole.
- **Hand on the horizon**: each finger at arm's length ≈ 15 min of sun before sunset.
- **Night**: stars turn 15°/hour round the Pole Star; the Plough works like a clock hand. Moon phases (below).
- Marked candles, sand-glasses, water clocks for intervals.

Calendar

- **Solstices** (sun rises at its most northern/southern point on the horizon — mark it with aligned stones/stakes): ~21 Jun and ~21 Dec. **Equinoxes** (sun rises due east, day = night): ~20 Mar and ~22 Sep.
- Year = 365¼ days → leap year every 4 years (except century years not divisible by 400). Keep a **tally stick** or written calendar, with a keeper and a copy.
- **Moon**: 29.5-day cycle. Full moon rises at sunset; a waxing moon is seen in the evening in the west (lit on the right in the northern hemisphere). Useful for counting months and tides (spring tides at new and full moon).
- **Day of the week** for any date: keep a printed perpetual calendar (Print pack).

Farming calendar (temperate northern hemisphere — adjust locally)

- **Jan–Feb**: plan, repair tools, prune fruit trees, sow under cover.
- **Mar–Apr**: sow peas, broad beans, onions, potatoes; prepare beds; lambing/kidding.
- **May–Jun**: after the last frosts, sow beans, squash, maize; shearing; haymaking.
- **Jul–Aug**: grain harvest; dry and store seed; preserving.
- **Sep–Oct**: lift potatoes, apples, nuts; sow winter wheat/rye; slaughter and salting.
- **Nov–Dec**: manure, store, crafts, teaching.
- Keep a **field diary**: dates of frosts, sowings, harvests and rain every year — in 5–10 years it gives a reliable local calendar.

28. Community and education — organising, deciding, teaching and preserving knowledge

Oracle OS guide. Manuals in the library: "Robert's Rules of Order", McGuffey Readers (reading and writing), Gutenberg (arithmetic, geometry), "50-year plan", Wikipedia/Wikibooks in Oracle.

A group's first 30 days

1. **Count people and skills**: list with name, age, health, trades (doctor, nurse, farmer, mechanic, teacher, cook, carpenter...).
2. **Common inventory**: water, food (days per person), medicines, tools, seeds, fuel.
3. **Priorities**: clean water → sanitation → shelter/heat → food → health → security → communication.
4. **Rotas**: water, firewood, cooking, watch, care of children/elderly/sick — everyone contributes according to ability.
5. **Simple written rules posted publicly**, approved by the assembly.

Making decisions

- Regular **assembly** (weekly) with an agenda, a chair and a secretary who records decisions in a book.
- Practical rule (simplified Robert's Rules): motion → someone seconds → time-limited debate where everyone speaks once before anyone speaks twice → vote → record. For serious matters, a 2/3 majority.
- Seek consensus in small groups; vote when there is none.
- **Rotate roles** with short terms (3–6 months); public accounting of the common inventory.
- **Resolving conflict**: first a talk between the parties; then a neutral mediator; then a council of 3–5 people chosen by lot or election. Proportionate, restorative sanctions (community work, restitution); expulsion is the last resort.
- Written basic rights: nobody is denied water, minimum food or medical care; children are protected; no corporal punishment; equality before the rules.

Economy without money

- Record of trades and hours worked (time bank), common store with goods in/out written down.
- Reserves for winter and bad harvests: keep 1 year of seed and, if possible, 1 year of grain.
- Agreed standard measures (weight, volume, length) kept in a public place.

Public health

- Latrines away from water, handwashing, boil/filter water, control rats and flies, isolate contagious patients (see Medicine and Natural building).
- Register of births, deaths, illnesses and vaccinations.

School

- Teach **everyone** to read, write and count — it is what preserves everything else. Classes by level, not age; advanced pupils teach the others.
- Minimum curriculum: reading/writing (McGuffey, primers), arithmetic and measures, practical geometry (areas, angles, 3-4-5 square), science (water, health, soil, plants, animals, simple chemistry), local history and geography, trades (apprenticeship with masters), first aid, ethics and living together.
- Materials: slates + chalk, wax tablets, charcoal, handmade paper; teach outdoors with sticks and stones (counting, fractions, geometry).
- **Apprenticeships**: each trade has masters and apprentices; the apprentice writes what they learn in the trade's notebook.
- Ceremonies and stories: songs, tales and festival calendars keep cohesion and memory.

Preserving knowledge

- Oracle is the library, but **paper is the backup**: print/hand-copy the Print pack and keep it in 2–3 dry, separate places.
- "Living library": each person responsible for studying and teaching 1–2 areas.
- Copyists: recopy decaying texts; keep a central index.
- Annual group history record (what happened, what was learned, harvests, weather).

29. Protecting family, food and shelter in a crisis — security without violence (English)

Oracle OS guide (Swedish, Norwegian, Finnish civil-protection guides, FEMA, Ready Scotland). The best defence is **not being a target**, **united neighbours** and **knowing when to leave**. Traps aimed at people are a crime almost everywhere and maim innocents (children, rescuers, neighbours) — don't use them.

Low profile

Don't show or post about food, water, fuel or generators; minimise light and noise at night if looting is reported; **split supplies** across several hiding places (and with relatives elsewhere); keep a small "giveaway" stash.

Harden the home

Locked doors and windows, quality lock/bar on the main door, film or bars on ground-floor windows, battery/solar motion lights, a dog. A **safe room** with a solid door, phone/radio, water, torch, first aid. Fire extinguisher and clear exits — fire is often a bigger risk than intruders.

Community — the strongest protection

Know neighbours; organise **street watch** shifts, signals (whistle, PMR446 channel), meeting point; share water, cooking, care for the elderly and children, radio news. Volunteer with **civil protection, fire services, Red Cross** or official civil defence — that is how to defend your community and country lawfully.

If someone threatens your family

Your life is worth more than food. Against weapons or groups, hand over what they want, observe, move away. Stay calm, hands visible, give them an exit. Children to the safe room; call **112 / 999 / 911** or use the radio; report afterwards. Self-defence must be **necessary and proportionate** to an ongoing attack; in the UK carrying weapons (including knives and sprays) for self-defence is illegal. Know your country's law.

Know when to leave

If the area becomes lawless, waterless or a combat zone, a **planned evacuation** is the best defence: grab bag, agreed route, meeting point outside town, discreet note for relatives.

Food from the land, legally

See the **small-game trapping** and fishing chapters of the SurvivalManual (US Army FM 21-76) in the library and follow local hunting laws.

30. Looking after your mind in a crisis (English)

Oracle OS guide (WHO Psychological First Aid). Fear, anxiety, sadness and irritability are **normal** reactions to abnormal events.

For yourself

Slow breathing (in for 4, out for 6, five times); 5-4-3-2-1 grounding (see, hear, touch, smell, taste); keep routines; limit news to set times; do one useful task a day; sleep, drink water, eat, move; avoid alcohol and drugs; talk to someone you trust.

Helping others — LOOK, LISTEN, LINK

Look: safety first, who needs urgent help. Listen: approach calmly, ask about needs, don't pressure people to tell their story, don't judge or make false promises. Link: water, food, shelter, information, family contact, services. Children: keep them with carers, explain simply and truthfully, let them play, keep routines.

Warning signs — get help now

Thoughts of death or self-harm, unable to care for oneself for days, severe confusion, hearing/seeing things others don't, uncontrolled alcohol or drug use. Emergency **112 / 999 / 911**; UK & Ireland **Samaritans 116 123**; USA & Canada **988**; Portugal **SNS 24 808 24 24 24**; Brazil **CVV 188**.

Oracle's chat companion can listen and help you calm down but is **not a psychologist or doctor**.

31. Stretching and pain — back, neck, arms, legs, feet, cramps and heavy effort

Oracle OS guide. Based on US Army FM 7-22 and ATP 7-22.02 (**illustrated** stretching and recovery drills), NIH/NIA "Exercise & Physical Activity", NHS (pain, cramps, sprains) and IFRC. For bleeding, tourniquets, fractures and wounds see also "First aid" and the illustrated "Stop the Bleed" PDF.

Red flags — get medical help (don't stretch it out)

- Back pain with **numbness around the genitals/buttocks**, loss of bladder or bowel control, or worsening leg weakness → **emergency** (nerve compression).
- Chest, left arm or jaw pain, breathlessness → could be the heart (999/112).
- **Swollen, warm, red, painful calf** → possible blood clot; don't massage.
- Pain after a fall or blow with deformity, can't bear weight, bone visible → fracture: immobilise.
- Pain with fever, unexplained weight loss, night pain that won't ease, or after cancer → medical assessment.

How to stretch well

1. **Warm up** 5 minutes first (walking, arm circles, marching on the spot). Cold muscle tears more easily.
2. Stretch to **comfortable tension**, never sharp pain. No bouncing.
3. **Hold 20–30 seconds**, 2–3 times each side. Breathe slowly, ease deeper on the out-breath.
4. Daily beats occasional. After a big effort, stretch gently and walk.
5. Pain that worsens during or the next day → do less.

Lower back

- **Knee to chest**: lying on your back, pull one knee to the chest, 20–30 s; switch; then both.
- **Pelvic tilt**: lying, knees bent, flatten the low back into the floor by tightening the tummy, 5 s, 10 times.
- **Cat–cow**: on all fours, round the back up (breathe out), then let the belly sink and look ahead (breathe in). 10 slow reps.
- **Child's pose**: kneel, sit back on the heels, arms stretched forward on the floor, forehead down. 30–60 s.
- **Lying rotation**: knees bent together, let them fall to one side, shoulders down, 20 s; other side.
- **Bridge**: lying, knees bent, lift the hips until knees-hips-shoulders line up, 5 s, 10 times (strong glutes protect the back).
- **Prone press-up** (disc-type pain that eases when standing): lie face down, prop on forearms 30 s; then on straight arms, hips down, 10 times. Stop if pain travels down the leg.

- **Figure-4 (piriformis / sciatic)**: lying, right ankle over left knee, pull the left thigh to the chest, 30 s; switch.
- **Hamstrings**: lying, leg straight up with a towel around the foot, 30 s.
- **Hip flexor**: half-kneeling, push the hips forward keeping the trunk upright, 30 s.
- In common low back pain, **staying active** (walking) recovers faster than bed rest. Heat eases stiffness.

Neck and shoulders

- **Chin tucks**: 5 s, 10 times — against "phone head".
- **Side bend**: ear to shoulder, the same-side hand pulls gently, 20 s each side.
- **Rotation**: look over one shoulder, then the other, slowly.
- **Upper trapezius**: hand behind the back, tilt the head the other way and slightly down, 20 s.
- **Shoulder rolls** backwards 10 times; **squeeze the shoulder blades** 5 s, 10 times.
- **Doorway chest stretch**: forearms on the door frame, step forward, 30 s.
- **Cross-body shoulder**: straight arm across the chest, pulled by the other arm, 20–30 s.
- **Pendulum** (sore shoulder): lean on a table and let the arm hang and make small circles, 1 minute.

Arms, elbows, wrists and hands

- **Wrist flexors**: arm straight, palm up, pull the fingers down with the other hand, 20–30 s.
- **Wrist extensors** (tennis elbow): arm straight, palm down, bend the wrist down, 20–30 s.
- **Triceps**: hand behind the head, the other hand pushes the elbow, 20 s.
- **Biceps**: hands clasped behind the back, arms straight, lift gently.
- **Tendon glides**: open hand, hook fist, full fist, tabletop — 10 times (tingling hands, repetitive work).
- **Tennis elbow** (pain on the outside of the elbow): slow strengthening — hold a light weight (bottle), palm down, lower the wrist over 3–4 s, 3×15 daily for 6–12 weeks.
- Thumb massage of the forearm relieves overload from carrying and gripping.

Legs

- **Quadriceps**: standing, supported, pull the heel to the bottom, knees together, 30 s.
- **Calves**: hands on a wall, back leg straight, heel down, 30 s; then with the back knee bent (soleus).
- **Adductors**: seated, soles together, knees down (butterfly), 30 s.
- **Glutes**: seated, cross one leg over the other and hug the knee to the chest, 30 s.

Feet

- **Plantar fasciitis** (heel pain on the first steps in the morning): calf stretches (above) several times a day; **towel stretch** around the sole pulling the toes toward you, 30 s, before getting up; **roll the sole** over a bottle (frozen helps), 2 minutes; supportive footwear, avoid barefoot on hard floors.
- **Toes**: pick up a towel with the toes; spread the toes; ankle circles.
- **Blisters**: don't pop if you can avoid it; protect with a plaster; if it bursts, wash, leave the skin, cover. Prevent: dry socks, well-laced boots, tape over friction spots.
- **Trench foot** (cold wet feet for many hours): dry, warm slowly, elevate, change socks daily.

Cramps

- **At the moment**: stretch the muscle the opposite way and hold until it eases — calf: straighten the knee and pull the toes toward you; front thigh: bend the knee pulling the heel up; sole: pull the toes up. Massage and walk a little.
- **Prevent**: drink through the day; in heat and hard work replace salts too (home ORS: 1 L water + 6 level teaspoons sugar + ½ level teaspoon salt); stretch calves before bed; warm up before effort.
- Frequent cramps, with weakness, swelling or after starting a new medicine → doctor.

After carrying someone or heavy loads for hours

- **Technique**: keep the load close, bend knees and hips (not the spine), back straight, lift with the legs, **don't twist** — turn your feet. Share the weight (2- or 4-person stretcher), switch sides every few minutes, rest every 10–15 minutes.
- **Moving casualties**: a clothing or blanket drag costs less than carrying in your arms; improvised stretcher (see "Search and rescue").
- **Recover**: walk slowly 5–10 min, stretch back, shoulders, forearms, quads and calves; drink and eat (protein + carbohydrate); sleep.
- **Delayed muscle soreness** (24–72 h later, stiff sore muscles): normal; gentle movement, heat, massage; no need for complete rest.
- **Muscle strain** (sudden pain during effort, sometimes bruising): first 2–3 days **protect** (offload), **elevate**, **compress** with a bandage, wrapped cold 15–20 min; then **move again** gradually and progressively without strong pain.

Bone and joint pain

- Pain at one spot on a bone, worse on weight-bearing after long marches → possible **stress fracture** (foot, shin): rest from impact 4–8 weeks.
- Sprain (twisted ankle): protect, elevate, compress, cold; weight-bear as tolerated; splint or figure-of-8 bandage.
- Arthritic knee or hip: regular gentle movement (walking, cycling, water) helps more than stopping.

Open wounds (summary)

- Firm direct pressure to stop bleeding (10 min). Life-threatening limb bleeding that won't stop: **tourniquet 5–7 cm above, tighten until it stops, write the time** (see "First aid" and "Stop the Bleed").
- Irrigate with plenty of clean water, remove dirt, cover with a clean dressing; change daily. Infection (spreading redness, heat, pus, fever) → antibiotics/doctor. Keep tetanus current.

Pain relief with and without medicines

- Heat (hot-water bottle, warm cloth) for stiffness and spasm; wrapped cold 15–20 min for a recent injury and swelling.
- Paracetamol (adult: 500 mg–1 g, max 4 g/day; less with liver disease or alcohol). Ibuprofen (adult: 200–400 mg with food, max 1.2 g/day without prescription) — **avoid** with ulcers, kidney disease, late pregnancy, blood thinners or asthma worsened by anti-inflammatories. Always check the leaflet.

32. Self-defence — Krav Maga, boxing, Muay Thai and jiu-jitsu (principles and core techniques)

Oracle OS guide. **Illustrated** manuals in the library: US Army TC 3-25.150 and FM 3-25.150 "Combatives" (based on Brazilian jiu-jitsu, plus boxing and Muay Thai), USMC "Marine Corps Martial Arts Program" and belt manuals (tan, grey, green), and classic boxing books. Nothing replaces training with an instructor and a partner: techniques are learned by controlled repetition.

Law and priorities

- Self-defence is only lawful if it is **necessary and proportionate** to stop an ongoing attack (UK: "reasonable force"; Portugal: self-defence, art. 32 of the Penal Code; Brazil: art. 25, moderate use). When the danger ends, you stop. This is not legal advice.
- Order of priority: **notice early → avoid → calm it down → escape → fight only to get away**. Winning means getting home.

1. Awareness and avoidance

- Look around, no earphones in risky areas; note exits, groups, hidden hands, someone closing in fast.
- Cross the street, go into a shop, draw attention. Money and phones can be replaced; you can't.
- **De-escalate**: calm voice, short phrases ("I don't want trouble. I'm leaving."), no insults, open hands in front of the chest as a "fence" (looks peaceful but protects the face and keeps distance).

2. Stance and guard (the base of every art)

- Feet shoulder-width, one forward (the weak side), knees slightly bent, weight on the balls of the feet.
- Hands at cheekbone height, elbows by the ribs, chin down, eyes on the opponent's chest (you see hands and feet).
- Movement: short steps, the foot nearest the direction moves first; never cross the legs.

3. Boxing — hands

- **Jab** (lead hand): fast, straight, turn the fist at the end, back to guard at once. Measures distance.
- **Cross** (rear hand): rotate the hip and rear foot, straight arm, shoulder protects the chin.
- **Hook**: arm bent at 90°, rotate the trunk; close range.
- **Uppercut**: upward, bending the knees.
- **Defences**: block with gloves/forearms, **slip** (tilt the head aside), **roll** (bend the knees and go under), step back or pivot out.

- Training combos: 1-2 (jab-cross), 1-2-3 (+ hook), jab-cross-step out diagonally.

4. Muay Thai — "the art of eight limbs"

- **Teep (push kick)**: lift the knee and push with the sole into the belly/hip — great for keeping distance and making space to escape.
- **Low roundhouse** to the thigh: pivot the standing foot, strike with the shin, same-side arm swings down for balance.
- **Knees**: at close range, pull the head/shoulders and drive the knee up.
- **Elbows**: short and powerful; very dangerous — only if essential.
- **Clinch**: hands behind the opponent's head, elbows together, controls their posture.
- **Checking kicks**: lift the shin on the side being kicked.

5. Krav Maga — street principles

- **Defend and counter at the same time**; short burst, then **escape**.
- Targets that stop an attack (only to escape): eyes, nose, throat, groin, knees.
- **Wrist grab**: rotate your arm toward the grabber's **thumb** (the weakest point) and pull hard.
- **Front choke** (hands on the neck): tuck the chin, bring hooked hands up inside and rip the thumbs away ("plucking"), counter (knee) and leave.
- **Rear arm choke**: turn the chin into their elbow, pull the arm down with both hands, drop your centre of gravity, turn in and get out.
- **Bear hug**: drop your weight (bend knees), widen the base, strike backward (heel to the foot, elbows), break free and run.
- **Knife or weapon**: hand over what they ask for and **run**. Knife defence is extremely risky even for experts; use barriers (chair, backpack, coat wrapped round the arm) only if there is no other way out.

6. Jiu-jitsu and ground fighting (the basis of "Army Combatives")

- **Falling safely**: chin to chest, slap the ground with the arms when falling backward, roll over the shoulder (not the head).
- **Technical stand-up**: seated, one hand on the ground behind, same-side foot in front to fend off, lift the hips and sweep the leg back under, stand already in guard.
- **Closed guard**: on your back with legs locked round their waist, you control them and limit strikes; pull them close, control the arms.
- **Hip escape ("shrimp")**: on your side, push off with the feet and move the hips away to make space and bring the legs back in front.
- **Bridge and roll ("upa")** when someone is mounted on you: trap one of their arms and the same-side foot, bridge the hips hard and roll to that side.
- **Side control**: frame the neck with the forearm, hip-escape and recover guard.
- **Chokes and locks** (rear naked choke, guillotine, armbar): in training, release at the first tap. A choke can cause unconsciousness in seconds and death; on the street, only as a last resort and release as soon as the threat ends.

- Street goal: **don't stay on the ground** — control, get up and leave (there may be more attackers, glass, objects).

7. Protecting others

- Put the child/person behind you and back toward an exit. Shout clear instructions ("Run into the shop!").
- In a group, nobody chases the attacker — everyone leaves together.

8. Training plan (3× a week, 45–60 min)

1. Warm-up 10 min (skipping, mobility — see "Stretching and pain").
2. Technique 15 min: guard, footwork, jab-cross, teep, breakfalls, shrimp, bridge.
3. Bag or pads 3×3 min (combinations) with 1 min rest.
4. Partner work with control 10 min: wrist releases, slow-motion choke defences, technical stand-up.
5. Conditioning 5–10 min: push-ups, squats, core, sprints.
6. Stretching and breathing.
7. Protection: hand wraps, gloves, gum shield; stop if joints hurt; never train strikes to the neck or head at full power.

9. After an attack

Get somewhere safe, check for injuries (see "First aid"), call 999/112, write down the attacker's description and witnesses, seek emotional support ("Mental health in a crisis").

33. Yoga — postures, breathing and relaxation (a safe 30–45 minute sequence)

Oracle OS guide. Based on the Government of India (Ministry of AYUSH) "Common Yoga Protocol" (in the library, **with photographs of every posture**), the Yoga Sutras of Patanjali and the Hatha Yoga Pradipika (classic texts, also in the library).

What it is

Yoga (Sanskrit *yuj*, "to join") is an Indian discipline over 2,000 years old. The **Yoga Sutras of Patanjali** (c. 2nd century BCE–5th century CE) describe 8 limbs: ethics (yama), discipline (niyama), posture (asana), breath (pranayama), withdrawal of the senses (pratyahara), concentration (dharana), meditation (dhyana) and absorption (samadhi). **Hatha Yoga** (Hatha Yoga Pradipika, 15th century) added postures, breath and energy techniques. Modern posture-based yoga developed mainly in the 20th century.

Safety

- Empty stomach (2–3 h after a meal). Mat or blanket, loose clothes.
- Never force: tension yes, pain no. Breathe through the nose; don't hold the breath in postures.
- **Pregnancy**: avoid strong twists, lying face down, Kapalabhati and inversions; prefer prenatal yoga.
- **High blood pressure, glaucoma, heart disease**: avoid inversions (head below heart) and Kapalabhati.
- **Disc herniation or sciatica**: avoid deep straight-legged forward bends; prefer gentle back-bends (Cobra) if they ease it.
- After surgery, with osteoporosis or injuries: adapt and seek guidance.

1. Prayer / intention (1 min)

Seated or standing, eyes closed, attention on the breath. Chant "Om" or set an intention.

2. Loosening (5 min)

- **Neck**: bend forward/back, sideways, and rotate gently, 4× each.
- **Shoulders**: hands on shoulders, elbow circles 5× forward and 5× back.
- **Trunk**: feet apart, twist side to side letting the arms swing, 5×.
- **Knees**: hands on knees, bend and straighten, 5×.

3. Standing postures

- **Tadasana (mountain)**: feet together, raise the arms breathing in, rise onto the toes, 10 s, lower breathing out.
- **Vrksasana (tree)**: sole of one foot on the inner thigh of the other leg (never on the knee), hands together at chest or overhead, gaze on one point, 20–30 s each side. Balance and focus.

- **Pada-hastasana (hands to feet)**: breathe in arms up, breathe out folding forward from the hips; knees may bend. Avoid with acute low back pain.
- **Ardha Chakrasana (half wheel)**: hands on the waist, breathe in and lean back, neck relaxed, 10 s.
- **Trikonasana (triangle)**: feet wide, arms out, bend sideways sliding a hand down the leg, other arm up, 20 s each side.

4. Seated postures

- **Bhadrasana (butterfly)**: soles together, knees down, back straight.
- **Vajrasana (diamond)**: sitting on the heels, back straight. Good after meals, for breathing and meditation.
- **Ardha Ustrasana (half camel)**: kneeling, hands on the waist, push the hips forward and lean back carefully, 10 s.
- **Sasankasana (hare/child)**: from Vajrasana, fold forward arms outstretched, forehead to the floor, 30 s. Relaxes the back.
- **Vakrasana (twist)**: seated, one leg crossed over the other, turn the trunk toward the bent knee, 20 s each side.

5. Face-down postures

- **Makarasana (crocodile)**: lie face down, forehead on crossed hands, relax. Eases the low back.
- **Bhujangasana (cobra)**: hands beside the chest, breathe in lifting the chest, elbows bent, hips down, 10–20 s. Helpful for low back pain that eases with extension.
- **Salabhasana (locust)**: arms by the sides, lift the straight legs (one at a time if hard), 10 s. Strengthens the back.

6. Lying on the back

- **Setu Bandhasana (bridge)**: knees bent, lift the hips, 10–20 s.
- **Uttanapadasana**: lift the straight legs to 30°, 10 s (bend the knees if the low back hurts).
- **Pavanamuktasana (wind-relieving)**: hug the knees to the chest, rock gently.
- **Shavasana (corpse)**: lying, arms apart, palms up, eyes closed; relax part by part from feet to head, 5–10 min. Essential: don't skip it.

7. Sun Salutation (Surya Namaskar) — 12 movements linked to the breath

1 Hands at the chest. 2. In: arms up, slight back-bend. 3. Out: fold forward, hands to the floor. 4. In: right leg back, knee down, look ahead. 5. Out: other leg back, plank. 6. Out: knees, chest and chin to the floor. 7. In: cobra. 8. Out: hips up (downward dog). 9. In: right foot forward between the hands. 10. Out: left foot forward, fold. 11. In: rise with arms up. 12. Out: hands to the chest. Repeat leading with the left leg. Start with 2–4 slow rounds.

8. Pranayama (breathing)

- **Belly breathing**: hand on the belly, breathe in through the nose and the belly rises, breathe out and it falls. The basis of everything.
- **Nadi Shodhana / Anulom Vilom (alternate nostril)**: right thumb closes the right nostril, breathe in left; ring finger closes the left, breathe out right; breathe in right; switch; breathe out left. 5–10 minutes. Calming and balancing.
- **Bhramari (bee)**: breathe in, and on the out-breath hum "mmm" like a bee with the ears closed. 5–10×. Excellent for anxiety and insomnia.
- **Sitali (cooling)**: curl the tongue into a tube, breathe in through it, out through the nose. For heat.
- **Ujjayi (ocean)**: gently narrow the throat so the breath sounds like the sea; slow rhythm.
- **Kapalabhati (skull-shining)**: short forceful out-breaths through the nose by pulling in the belly, passive in-breath; 30 × 3 rounds. **Avoid** in pregnancy, high blood pressure, heart disease, hernia, epilepsy and menstruation.
- Long breath retentions (kumbhaka) and bandhas: only with a teacher.

9. Meditation and closing

Sit upright 5–10 min watching the breath (see the "Meditation" guide). End with an intention (sankalpa) and a few seconds of gratitude.

Short sequences for real situations

- **Stiff low back**: cat–cow, child, crocodile, cobra, bridge, lying rotation, shavasana.
- **Before sleep**: legs up the wall (lying with the legs resting on a wall, 5 min), reclined butterfly, Bhramari, shavasana.
- **Anxiety/fear**: belly breathing with a longer out-breath than in-breath (4 in, 6–8 out), Bhramari, alternate nostril.
- **After carrying loads**: child, cat–cow, seated twist, cross-body shoulder, forearm stretches.

34. Meditation — every major type, from ancient traditions to modern science

Oracle OS guide. Source texts in the library: Upanishads, Bhagavad Gita, Yoga Sutras of Patanjali, Hatha Yoga Pradipika, Dhammapada, "With Each & Every Breath" (Thanissaro), "Mindfulness in Plain English" (Gunaratana), Tao Te Ching, Marcus Aurelius' Meditations, "The Practice of the Presence of God" (Brother Lawrence), "The Imitation of Christ", WHO "Doing What Matters in Times of Stress" and US Army FM 7-22 (sleep, breathing and mind).

Getting started (applies to almost all)

- **Posture**: seated with the back upright but not rigid (chair, cushion or kneeling), hands resting, chin slightly tucked. You can also meditate lying down (for sleep) or walking.
- **Time**: 5–10 minutes daily beats an hour now and then. Build up slowly.
- **The mind will wander**: that's normal. Meditation is precisely noticing and **coming back**, without self-criticism.
- **Caution**: meditation can bring up difficult emotions and memories. With trauma, severe depression, psychosis or panic attacks: short sessions, eyes open, focus on the body and surroundings, and support from someone you trust or a professional. If distress grows, stop and do something grounding (walk, cold water on the hands, talk to someone).

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1. Ancient India — Vedic and yogic traditions (c. 1500 BCE onward)

- **Mantra / Japa**: repeat a sacred word or phrase (Om, "So Ham", "Om Namah Shivaya", a name of God) aloud, whispered or silently; traditionally 108 times on a mala (bead rosary). When the mind wanders, return to the sound.
- **So-Ham / Hamsa**: silently hear "So" on the in-breath and "Ham" on the out-breath ("I am That"). Joins mantra and breath.
- **Self-inquiry (Atma vichara)** — Upanishads; Ramana Maharshi (20th century): ask "Who am I?" and trace the source of the "I" thought without accepting verbal answers.
- **Dharana → Dhyana → Samadhi** (Yoga Sutras): fix attention on an object (flame, image, point between the brows, breath), sustain it continuously (meditation) until the split between observer and object dissolves (absorption).
- **Trataka (steady gazing)** — Hatha Yoga: gaze at a candle flame 1 m away without blinking until the eyes water, then close them and hold the inner image. Avoid with epilepsy.
- **Nada Yoga (inner sound)**: close the ears and listen to inner sounds (humming, bells, flute), following the subtlest.
- **Pranayama as meditation**: alternate-nostril breathing, Bhramari (see "Yoga").

- **Yoga Nidra (yogic sleep)**: lying down, rotate awareness through the body, notice opposite sensations (warm/cold), images and an intention (sankalpa). Rooted in ancient tantric practices (nyasa); the modern form was systematised by Swami Satyananda in the 1960s. Excellent for insomnia and exhaustion.
- **Chakra and Kundalini meditation** (tantric traditions, 8th–16th centuries): see "Chakras".
- **Bhakti (devotion)**: chanting (kirtan), visualising and offering the heart to a form of the divine.

2. Jainism

- **Samayika**: 48 minutes of equanimity — sit, renounce attachment and aversion, observe the soul as pure witness.
- **Preksha**: perception of breath, body and inner colours (modern form).

3. Buddhism (from the 5th century BCE)

- **Samatha (calm)**: concentration on one object until the mind is stable and serene.
- **Anapanasati (mindfulness of breathing)** — the Buddha; 16 steps in 4 tetrads: body (long breath, short breath, whole body, calming), feelings (rapture, ease), mind (knowing, gladdening, steadying, releasing the mind) and wisdom (impermanence, fading, cessation, letting go). Simple practice: feel the air at the nostrils or the belly; count 1 to 10 and restart if you lose count.
- **Satipatthana (4 foundations of mindfulness)**: observe body, feelings (pleasant/unpleasant/neutral), mind states and phenomena (hindrances, senses, factors of awakening).
- **Vipassana (insight)**:
  - *Noting* (Mahasi Sayadaw style, Burma): silently note "rising, falling" of the belly and label what arises: "thinking", "hearing", "pain"…
  - *Body sweeping* (U Ba Khin / S. N. Goenka): move attention from head to feet observing sensations without reacting, seeing that they change.
- **Metta (loving-kindness)** and the 4 **Brahmaviharas**: wish "May I be safe, healthy, at peace, may I live with ease", then for a loved one, a neutral person, a difficult person and all beings. Then **karuna** (compassion), **mudita** (joy in others' happiness) and **upekkha** (equanimity).
- **Recollections (anussati)**: of the qualities of the Buddha, the teaching, the community, generosity; **maranasati** (contemplation of death) to value time.
- **Walking meditation (cankama / kinhin)**: walk slowly 15–20 steps, feeling lift, move, place of each foot; turn and return.
- **Zen** (China/Japan, 6th–13th centuries):
  - *Zazen / Shikantaza* ("just sitting", Soto school): eyes half-open resting on the floor 1 m ahead, firm posture, let thoughts pass with no object.
  - *Koan* (Rinzai school): hold a paradoxical question ("What was your face before your parents were born?") until habitual thinking breaks; traditionally with a teacher.
- **Tibetan Buddhism**:

- *Shamatha with an object* (stone, image, breath).
- *Tonglen* (giving and taking): breathing in, imagine taking in someone's suffering as dark smoke; breathing out, send relief as light. Builds compassion without avoiding pain.
- *Deity visualisation and mantras* (e.g. "Om Mani Padme Hum" of Avalokiteshvara) — Vajrayana practices traditionally transmitted by a teacher.
- *Mahamudra / Dzogchen*: resting in the nature of awareness itself — requires guidance.

- **Pure Land**: reciting the name of Amitabha Buddha ("Namo Amituofo" / "Namu Amida Butsu") with faith and attention.

4. China — Taoism and internal arts

- **Zuowang ("sitting and forgetting")** — Zhuangzi: let go of perception, knowledge and self until one with the Tao.
- **Shouyi ("guarding the One")**: holding attention at the centre of body and spirit.
- **Dantian breathing**: attention three fingers below the navel, slow low breathing; basis of Qigong and internal martial arts.
- **Microcosmic orbit**: guide attention with the breath up the spine to the crown and down the front of the body to the belly.
- **Zhan Zhuang ("standing like a tree")**: standing, knees softly bent, arms as if hugging a tree, 5–20 minutes still and relaxed.
- **Tai Chi and Qigong**: slow moving meditation.

5. Greece and Rome — Stoicism and philosophers

- **Evening review** (Seneca, Pythagoras): "What did I do well? Where did I fall short? What will I do differently?"
- **Premeditation of adversity** (premeditatio malorum): calmly imagine possible losses so you can face them with composure and appreciate the present.
- **View from above** (Marcus Aurelius): imagine rising to see the city, the country, the Earth — putting problems in perspective.
- **Dichotomy of control** (Epictetus): separate what is up to you (judgements, actions) from what is not, and spend energy only on the first.

6. Christian traditions

- **Jesus Prayer / Hesychasm** (Desert Fathers, 4th century; Philokalia): repeat "Lord Jesus Christ, Son of God, have mercy on me" with the breath, bringing attention "from the head into the heart".
- **Lectio Divina** (St Benedict, 6th century): read a Bible passage slowly (lectio), chew on one phrase (meditatio), respond in prayer (oratio), rest in silence (contemplatio). The Bible in English and Portuguese is in the library.
- **Ignatian contemplation and the Examen** (St Ignatius, 16th century): imagine yourself inside a Gospel scene with all five senses; at night, review the day with gratitude.

- **Rosary**: meditative repetition of prayers while contemplating episodes of Christ's life.
- **Centering Prayer** (1970s, inspired by "The Cloud of Unknowing", 14th century): choose a simple sacred word and, whenever a thought arises, gently return to it; 20 minutes.
- **Practice of the presence of God** (Brother Lawrence, 17th century): keep an inner conversation with God during ordinary tasks.
- **Taizé chant**: short phrases sung repeatedly in a group.

7. Jewish traditions

- **Hitbodedut** (Rabbi Nachman of Breslov, 18th century): speak alone with God, quietly, in nature, in your own words, daily.
- **Contemplative recitation of the Psalms**; **Shabbat** as mindful rest; Kabbalistic meditations on divine names (traditionally with a teacher).

8. Islam and Sufism

- **Dhikr (remembrance of God)**: repeat "La ilaha illallah", "Allah", or "Subhanallah, Alhamdulillah, Allahu Akbar" (33 times each, on a tasbih), with attention in the heart.
- **Muraqaba (watchfulness)**: sit in silence aware of God's presence.
- **Tafakkur (contemplation)**: reflect on nature and creation.
- **Sema** (Mevlevi whirling, the tradition of Rumi): prayer in movement, learned in community.

9. Indigenous and nature traditions

Across many peoples: contemplative silence in nature, rhythmic drumming, chant, watching the sky and the cycles. An accessible practice: sit 20 minutes in the same outdoor spot every day, only listening and watching.

10. Modern and scientific methods (20th–21st centuries)

- **Progressive muscle relaxation** (Jacobson, 1920s): tense 5 s and release 15 s each muscle group — feet, legs, buttocks, belly, hands, arms, shoulders, face.
- **Autogenic training** (Schultz, 1932): silently repeat "my right arm is heavy… warm…", then whole body, calm heart, easy breathing.
- **Relaxation response** (Benson, 1975): repeated word + passive attitude, 10–20 minutes.
- **Transcendental Meditation** (1950s): personal mantra 20 minutes twice a day (taught by instructors).
- **MBSR — Mindfulness-Based Stress Reduction** (Kabat-Zinn, 1979) and **MBCT** (to prevent depression relapse): body scan, breathing, gentle yoga, everyday mindfulness.
- **Body scan**: lying down, move attention from the toes to the top of the head, noticing sensations without changing them, 10–40 minutes.

- **Open monitoring**: no fixed object; notice whatever arises (sounds, thoughts, sensations) and let it pass.
- **Visualisation / guided imagery**: imagine a safe place in detail (sounds, smells, temperature).
- **Gratitude**: write down 3 good things from the day and why.

11. Breathing to calm the body now

- **Box (4-4-4-4)**: in 4 s, hold 4, out 4, hold 4; 4 cycles. Used by soldiers and first responders.
- **4-7-8**: in 4, hold 7, out 8 through the mouth; 4 cycles (sleep).
- **Coherent breathing**: ~6 breaths per minute (5 s in, 5 s out) for 5–10 minutes.
- **Physiological sigh**: two short in-breaths through the nose followed by one long out-breath through the mouth; 1–3 times.
- **Long exhale**: always breathe out longer than in (e.g. 4 in, 6–8 out).

12. Short practices for emergencies and disasters

- **5-4-3-2-1 grounding** (panic, flashbacks): name 5 things you see, 4 you touch, 3 you hear, 2 you smell, 1 you taste.
- **STOP**: Stop, Take a breath, Observe (body, emotion, situation), Proceed with the most useful action.
- **Hands on the chest**: one hand on the heart and one on the belly, feel the warmth, tell yourself "this is really hard; I am doing the best I can".
- **Grief**: light a candle, say the name of the one lost, breathe with the pain without pushing it away; share with someone.
- **To sleep in a shelter**: lying body scan + 4-7-8 breathing or Yoga Nidra.
- To support others in crisis, see "WHO — Psychological first aid" and the "Mental health in a crisis" guide.

35. Chakras — origins, the seven centres, nadis, Kundalini and how to meditate with them

Oracle OS guide. Based on the Hatha Yoga Pradipika and the Yoga Sutras (in the library), the *Ṣaṭ-cakra-nirūpaṇa* (Pūrṇānanda, 1577) translated by Sir John Woodroffe ("Arthur Avalon") in *The Serpent Power* (1919), and historical scholarship on tantra.

What they are — and what they are not

- *Cakra* is Sanskrit for "wheel" or "circle". In Hindu and Buddhist tantric texts (8th–16th centuries) they are **centres of concentration** visualised along the body's axis, used as a map for meditation, mantra and visualisation.
- **The number varies** by tradition: tantric Buddhism usually has 4 or 5; several Hindu texts give 6 + 1 (the best known today, from the *Ṣaṭ-cakra-nirūpaṇa*); others give more.
- The **rainbow colours** (red → violet) linked to 7 chakras are a 20th-century Western creation; the old texts describe different colours for each lotus.
- There is no scientific evidence that chakras are physical structures or that they cure disease. As an **attention and relaxation practice** they are safe and can calm. They don't replace medical treatment.

The channels (nadis) and Kundalini

- **Sushumna**: central channel along the spine.
- **Ida** (left, lunar, calming) and **Pingala** (right, solar, activating) twist around Sushumna. Alternate-nostril breathing (Nadi Shodhana) aims to balance them.
- **Kundalini**: energy described as a sleeping serpent coiled at the base of the spine; in tantric practice it awakens and rises through Sushumna, piercing the chakras up to the crown.

The seven chakras (the most widespread Hindu system)

Chakra	Location	Element	Bija (seed sound)	Petals	Colour in old texts	Modern colour	Traditional/ modern themes
Muladhara (root)	base of spine / perineum	earth	LAM	4	yellow square, crimson petals	red	stability, survival, safety
Svadhisthana (sacral)	lower belly / genitals	water	VAM	6	white crescent, vermilion petals	orange	creativity, emotion, sexuality
Manipura (navel / solar plexus)	navel	fire	RAM	10	red triangle, dark blue petals	yellow	will, energy, digestion
Anahata (heart)	centre of chest	air	YAM	12	smoky hexagram, bright red petals	green	love, compassion, connection

Chakra	Location	Element	Bija (seed sound)	Petals	Colour in old texts	Modern colour	Traditional/ modern themes
Vishuddha (throat)	throat	ether / space	HAM	16	white circle, smoky purple petals	blue	expression, truth, com munication
Ajna (third eye)	between the brows	mind	OM	2	white	indigo	intuition, dis cernment
Sahasrara (crown)	top of the head	beyond the elements	(silence / OM)	1000	brilliant white	violet / white	unity, awareness

Similar systems elsewhere

- **Tibetan Buddhism**: channels (tsa), winds (lung) and drops (tigle); chakras at the crown, throat, heart and navel (sometimes the secret place), with the syllables OM, AH, HUM.
- **Taoism**: three **dantian** — lower (below the navel, vital energy), middle (heart, emotions) and upper (between the brows, spirit); the microcosmic orbit.
- **Sufism**: *lataif* (subtleties) at points in the chest and head, worked with dhikr.

How to meditate with the chakras (20–30 minutes)

1. Sit upright, eyes closed, 10 slow belly breaths.
2. Bring attention to the **base of the spine**. Feel contact with the ground. If you like, chant "LAM" 3–7 times (or silently) feeling the vibration there. Imagine a light (red, or any colour you prefer) and the quality of **safety and stability**.
3. Move up to the **lower belly** — "VAM", flow. Then the **navel** — "RAM", calm strength. The **heart** — "YAM", tenderness and compassion (you can add loving-kindness). The **throat** — "HAM", true speech. **Between the brows** — "OM", clarity. The **crown** — silence, openness.
4. 1–3 minutes at each centre. At the end, imagine light running up the spine and back down, and rest in silence.
5. **Close by coming down**: bring attention back to the feet and the ground, move the hands, open the eyes. This prevents feeling "light-headed".

Short versions

- **Spinal breath**: breathing in, feel attention rise from base to crown; breathing out, descend. 5 minutes.
- **Heart only**: hand on the chest, breathe into the centre of the chest, repeat "YAM" or a loving-kindness phrase.
- **Rooting** in moments of fear: feet on the ground, weight in the legs, slow breath, attention at the base of the body.

Cautions

- Intense Kundalini practices (long breath retention, strong bandhas, hours of concentration) can cause dizziness, insomnia, anxiety, heat or uncontrolled energy sensations ("Kundalini crisis"). Do them only with an experienced teacher.

- Avoid breath retention with high blood pressure, heart disease, pregnancy, epilepsy or a history of psychosis.
- If physical symptoms appear (chest pain, breathlessness, numbness), treat them as medical, not as "energy".

Perpetual calendar (2000–2100)

Find the year in the table: the number tells you which of the 14 calendars to use. Calendars 8–14 are leap years.

2000: **13**	2001: **1**	2002: **2**	2003: **3**	2004: **11**	2005: **6**	2006: **7**	2007: **1**
2008: **9**	2009: **4**	2010: **5**	2011: **6**	2012: **14**	2013: **2**	2014: **3**	2015: **4**
2016: **12**	2017: **7**	2018: **1**	2019: **2**	2020: **10**	2021: **5**	2022: **6**	2023: **7**
2024: **8**	2025: **3**	2026: **4**	2027: **5**	2028: **13**	2029: **1**	2030: **2**	2031: **3**
2032: **11**	2033: **6**	2034: **7**	2035: **1**	2036: **9**	2037: **4**	2038: **5**	2039: **6**
2040: **14**	2041: **2**	2042: **3**	2043: **4**	2044: **12**	2045: **7**	2046: **1**	2047: **2**
2048: **10**	2049: **5**	2050: **6**	2051: **7**	2052: **8**	2053: **3**	2054: **4**	2055: **5**
2056: **13**	2057: **1**	2058: **2**	2059: **3**	2060: **11**	2061: **6**	2062: **7**	2063: **1**
2064: **9**	2065: **4**	2066: **5**	2067: **6**	2068: **14**	2069: **2**	2070: **3**	2071: **4**
2072: **12**	2073: **7**	2074: **1**	2075: **2**	2076: **10**	2077: **5**	2078: **6**	2079: **7**
2080: **8**	2081: **3**	2082: **4**	2083: **5**	2084: **13**	2085: **1**	2086: **2**	2087: **3**
2088: **11**	2089: **6**	2090: **7**	2091: **1**	2092: **9**	2093: **4**	2094: **5**	2095: **6**
2096: **14**	2097: **2**	2098: **3**	2099: **4**	2100: **5**			

Calendar 1

January

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

February

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

March

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

April

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

May

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

June

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

July

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

August

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

September

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

October

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

November

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

December

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

Calendar 2

January

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

February

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28			

March

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

April

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

May

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

June

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

July

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

August

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

September

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

October

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

November

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

December

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Calendar 3

January

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

February

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28		

March

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

April

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

May

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

June

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

July

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

August

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

September

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

October

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

November

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

December

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Calendar 4

January

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

February

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	

March

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

April

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

May

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

June

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

July

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

August

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

September

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

October

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

November

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

December

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Calendar 5

January

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

February

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

March

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

April

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

May

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

June

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

July

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

August

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

September

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

October

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

November

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

December

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Calendar 6

January

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

February

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28						

March

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

April

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

May

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

June

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

July

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

August

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

September

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

October

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

November

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

December

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Calendar 7

January

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

February

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28					

March

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

April

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

May

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

June

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

July

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

August

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

September

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

October

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

November

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

December

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Calendar 8 (bissexto / leap)

January

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

February

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

March

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

April

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

May

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

June

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

July

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

August

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

September

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

October

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

November

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

December

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Calendar 9 (bissexto / leap)

January

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

February

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29		

March

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

April

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

May

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

June

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

July

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

August

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

September

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

October

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

November

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

December

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Calendar 10 (bissexto / leap)

January

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

February

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	

March

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

April

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

May

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

June

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

July

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

August

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

September

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

October

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

November

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

December

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Calendar 11 (bissexto / leap)

January

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

February

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

March

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

April

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

May

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

June

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

July

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

August

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

September

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

October

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

November

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

December

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Calendar 12 (bissexto / leap)

January

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

February

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29						

March

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

April

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

May

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

June

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

July

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

August

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

September

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

October

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

November

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

December

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Calendar 13 (bissexto / leap)

January

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

February

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29					

March

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

April

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

May

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

June

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

July

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

August

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

September

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

October

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

November

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

December

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Calendar 14 (bissexto / leap)

January

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

February

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29				

March

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

April

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

May

M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

June

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

July

M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

August

M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

September

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

October

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

November

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

December

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						